



STATE OF CALIFORNIA
DEPARTMENT OF TRANSPORTATION

**NOTICE TO BIDDERS
AND
SPECIAL PROVISIONS**

**FOR CONSTRUCTION ON STATE HIGHWAY AND CITY STREET IN MARIN
COUNTY IN SAN RAFAEL AT FREITAS PARKWAY OVERCROSSING**

In District 04 On Route 101

Under

Bid book dated September 30, 2024

Standard Specifications dated 2023

Project plans approved January 24, 2024

Standard Plans dated 2023

Identified by

Contract No. 04-0K8004

04-Mrn-101-13.7

Project ID 0416000141

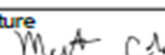
Federal-Aid Project

ACNH-Q101(428)E

SPECIAL NOTICES

- See sections 2 and 3 for contractors' registration requirements.
- The Department advises bidders that potential claim records must be submitted by the contractor using the Department's Internet potential claim system.
- See section 2 for submittal requirements for DBE quotes, DVBE quotes, and Non–Small Business Subcontractor Preference.
- For work plan for local material from (1) a noncommercial source or (2) a source not regulated under California jurisdiction, see section 6-1.03B(1).
- See section 7-1.02K(3) for the requirements for electronic submittal of certified payroll records using LCPtracker Pro.
- The flagging and temporary traffic control requirements have been revised. See sections 7-1.03, 7-1.04, and 12.
- See section 14-11.14 for changes to the management of treated wood waste.
- The schedules for the submittal of DBE forms have been revised. See section 2-1.33 for the submittal schedules.
- See section 2-1.04 for mandatory prebid meeting requirements.

CONTRACT No. 04-0K8004

DESIGN OVERSIGHT APPROVAL		REGISTRATION No.	DATE
Printed Name MERLITO MARK COLOMA	Signature 	76602	8/1/24

Approved as to impact on State facilities and conformance with applicable State standards and practices and that technical oversight was performed as described in the California Department of Transportation A & E Consultant Services Manual.

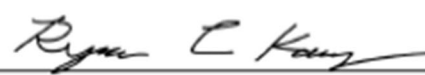
The special provisions contained herein have been prepared by or under the direction of the following Registered/Licensed Professionals.

ROADWAY


REGISTERED CIVIL ENGINEER 7/30/2024 DATE



ELECTRICAL


REGISTERED CIVIL ENGINEER 07/29/2024 DATE



STRUCTURES


REGISTERED CIVIL ENGINEER 11/16/2023 DATE



LANDSCAPE



LICENSED LANDSCAPE ARCHITECT



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STANDARD PLANS LIST

The standard plan sheets applicable to this Contract include those listed below. The applicable revised standard plans (RSPs) listed below are included in the project plans.

A3A	Abbreviations (Sheet 1 of 3)
A3B	Abbreviations (Sheet 2 of 3)
A3C	Abbreviations (Sheet 3 of 3)
A10A	Legend - Lines and Symbols (Sheet 1 of 5)
A10B	Legend - Lines and Symbols (Sheet 2 of 5)
A10C	Legend - Lines and Symbols (Sheet 3 of 5)
A10D	Legend - Lines and Symbols (Sheet 4 of 5)
A10E	Legend - Lines and Symbols (Sheet 5 of 5)
A10F	Legend - Soil (Sheet 1 of 2)
A10G	Legend - Soil (Sheet 2 of 2)
A10H	Legend - Rock
A20A	Pavement Markers and Traffic Lines - Typical Details
RSP A20B	Pavement Markers and Traffic Lines - Typical Details
A20C	Pavement Markers and Traffic Lines - Typical Details
A20D	Pavement Markers and Traffic Lines - Typical Details
A20E	Traffic Lines - Typical Details for Contrast Striping
A20F	Pavement Markers and Traffic Lines - Typical Details
A20G	Exit Ramp with Enhanced Pavement Markers for Wrong Way Details
A24A	Pavement Markings - Arrows
A24B	Pavement Markings - Arrows and Symbols
A24C	Pavement Markings - Symbols and Numerals
A24D	Pavement Markings - Words
A24E	Pavement Markings - Words
A24F	Pavement Markings - Crosswalks
A24G	Pavement Markings - Yield Lines, Limit Lines, and Wrong Way Details
A24H	Pavement Markings - Arrows
A62A	Excavation and Backfill - Miscellaneous Details
A62B	Limits of Payment for Excavation and Backfill - Bridge Surcharge and Wall
A62D	Excavation and Backfill - Concrete Pipe Culverts
A62DA	Excavation and Backfill - Concrete Pipe Culverts - Indirect Design Method
A62F	Excavation and Backfill - Metal and Plastic Culverts

A63A	Portable Concrete Barrier (Type 60K)
A63B	Portable Concrete Barrier (Type 60K)
A73A	Object Markers
A73B	Markers
A73C	Delineators, Channelizers and Barricades
A74	Survey Monuments
A74A	Global Navigation Satellite System Station Details
A85	Chain Link Fence
A85A	Chain Link Fence Details
A85B	Chain Link Fence Details
A87A	Curbs and Driveways
A87B	Hot Mix Asphalt Dikes
A88A	Curb Ramp Details
A88B	Curb Ramp and Island Passageway Details
P1	Jointed Plain Concrete Pavement - New Construction
P10	Concrete Pavement Dowel Bar Details
P12	Concrete Pavement Dowel Bar Basket Details
P15	Concrete Pavement - Tie Bar Details
P17	Concrete Pavement - Tie Bar Basket Details
D71	Drainage Inlet Markers
D72B	CIP Drainage Inlets - Types G1, G2, G3, G4, G5 and G6
D72C	CIP Drainage Inlets - Types G1, G2, G3, G4, G5 and G6
D72E	CIP Drainage Inlets - Types GO and GDO
D72F	CIP Drainage Inlet Notes
D72G	CIP Drainage Inlet Tables
D73B	Precast Drainage Inlets - Types G1, G2, G3, G4, G5 and G6
D73C	Precast Drainage Inlets - Types G1, G2, G3, G4, G5 and G6
D73E	Precast Drainage Inlets - Types GO and GDO
D73F	Precast Drainage Inlet Notes
D73G	Precast Drainage Inlet Tables
D74	Drainage Inlet Details
D77A	Grate Details No. 1
D77B	Grate Details No. 2
D78A	Gutter Depressions
D87D	Overside Drains

D89A	Pipe Culvert Headwalls - Straight and "L"
D89B	Pipe Culvert Headwalls - Straight and "L"
H1	Landscape and Erosion Control Symbols
H2	Landscape Details
H3	Landscape Details
H4	Landscape Details (Sprinkler Assembly)
RSP H5	Landscape Details (Swing Joint and Protector)
H6	Landscape Details
H7	Landscape Details
H8	Landscape Details
H9	Landscape Details
H10	Irrigation Controller Enclosure Cabinet
H51	Erosion Control Details - Fiber Roll and Compost Sock
H52	Rolled Erosion Control Product
RSP T1A	Temporary Crash Cushion, Sand Filled (Unidirectional)
T1A1	Temporary Crash Cushion, Sand Filled (Unidirectional)
T1B	Temporary Crash Cushion, Sand Filled (Bidirectional)
T2	Temporary Crash Cushion, Sand Filled (Shoulder Installations)
T3A	Temporary Railing (Type K)
T3B	Temporary Railing (Type K)
T3C	Temporary Barrier System (Cross Bolt)
T3D	Temporary Barrier System (Cross Bolt)
T3E	Temporary Barrier System (Cross Bolt)
T9	Traffic Control System Tables for Lane and Ramp Closures
T14	Traffic Control System for Ramp Closure
T30	Temporary Pedestrian Access Routes - Typical Sidewalk Closure and Pedestrian Detour
T31	Temporary Pedestrian Access Routes - Typical Sidewalk Diversion Within Roadbed
T32	Temporary Pedestrian Access Routes - Typical Sidewalk/Crosswalk Closure and Pedestrian Detour
T33	Temporary Pedestrian Access Routes - Ramp
T34	Temporary Pedestrian Access Routes - Curb Ramp Options
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T59	Temporary Water Pollution Control Details (Temporary Concrete Washout Facility)
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T63	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T64	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T65	Temporary Water Pollution Control Details (Temporary High-Visibility Fence)
B0-1	Bridge Details
B0-3	Bridge Details
B3-3B	Retaining Wall Type 1A (Case 2)
B3-5	Retaining Wall Details No. 1
B3-6	Retaining Wall Details No. 2
B7-11	Utility Details
RS1	Roadside Signs - Typical Installation Details No. 1
RS2	Roadside Signs - Wood Post - Typical Installation Details No. 2
RS4	Roadside Signs - Typical Installation Details No. 4
ES-1A	Electrical Systems (Legend)
ES-1B	Electrical Systems (Legend)
RSP ES-1C	Electrical Systems (Legend)
ES-2A	Electrical Systems (Service Equipment)
ES-2D	Electrical Systems (Service Equipment Enclosure and Typical Wiring Diagram, Type III - A Series)
ES-3C	Electrical Systems (Controller Cabinet Foundation and Pad Details)
ES-3H	Electrical Systems (Irrigation Controller Enclosure Cabinet)
ES-3K	Electrical Systems (Electronics Assembly Connection Diagram) (Battery Backup System)
ES-4A	Electrical Systems (Signal Head Mounting)
ES-4B	Electrical Systems (Pedestrian Signal Heads)
ES-4C	Electrical Systems (Signal Heads and Mountings)
ES-4D	Electrical Systems (Signal Head Mounting)
RSP ES-4E	Electrical Systems (Signal Heads)
ES-5A	Electrical Systems (Loop Detectors)
ES-5B	Electrical Systems (Detectors)
ES-5C	Electrical Systems (Accessible Pedestrian Signal and Push Button Assemblies)
ES-5D	Electrical Systems (Curb and Shoulder Termination, Trench, and Handhole Details)
RSP ES-5E	Electrical Systems (Trench in Pavement)
ES-6A	Electrical Systems (Lighting Standard, Types 15 and 21)
ES-6B	Electrical Systems (Electrolier Anchorage and Grouting for Type 15 and Type 21 Barrier Rail Mounted)
ES-7A	Electrical Systems (Signal and Lighting Standard, Type TS, and Type 2 Post)
RSP ES-7B	Electrical Systems (Signal and Lighting Standard, Type 1 and Equipment Identification Characters)

RSP ES-7J	Electrical Systems (Flashing Beacon on a Type 1, Type 15-FBS and Type 40 Standard)
RSP ES-7M	Electrical Systems (Signal and Lighting Standard, Detail No. 1)
ES-7N	Electrical Systems (Signal and Lighting Standard, Detail No. 2)
ES-7O	Electrical Systems (Signal and Lighting Standard, Detail No. 3)
ES-7Q	Electrical Systems (Pedestrian Barricades)
ES-8A	Electrical Systems (Non-Traffic Pull Box)
ES-11	Electrical Systems (Foundation Installations)
ES-13A	Electrical Systems (Splice Insulation Methods Details)
ES-13B	Electrical Systems (Kinking and Banding Detail)
ES-14B	Electrical Systems (EMS and FBCA Wiring Diagrams)

CANCELED STANDARD PLANS LIST

The standard plan sheets listed below are canceled and not applicable to this contract.

Plan No.	Date Canceled	Plan No.	Date Canceled	Plan No.	Date Canceled
A77U1	01-22-24				
A77U2	01-22-24				
A77U3A	01-22-24				
A77U3B	01-22-24				
A77U4	01-22-24				
A77U5	01-22-24				
A77V1	01-22-24				
A77V2	01-22-24				
A78A	01-22-24				
A78B	01-22-24				
A78C1	01-22-24				
A78C2	01-22-24				
A78D1	01-22-24				
A78D2	01-22-24				
A78E1	01-22-24				
A78E2	01-22-24				
A78E3	01-22-24				
A78F1	01-22-24				
A78F2	01-22-24				
A78GA	01-22-24				
A78GB	01-22-24				
A78H	01-22-24				
A78J	01-22-24				
A78K	01-22-24				

NOTICE TO BIDDERS

Bids open Wednesday, November 20, 2024

Dated September 30, 2024

General work description: Construct roundabout, curb ramps and SW, signal and lighting systems.

The Department will receive sealed bids for CONSTRUCTION ON STATE HIGHWAY AND CITY STREET IN MARIN COUNTY IN SAN RAFAEL AT FREITAS PARKWAY OVERCROSSING.

District-County-Route-Post Mile: 04-Mrn-101-13.7

Contract No. 04-0K8004

The Contractor must have either a Class A license or any combination of the following Class C licenses which constitutes a majority of the work: C-12, C-8.

The DBE Contract goal is 22 percent.

Federal-aid project no.:

ACNH-Q101(428)E

Bids must be on a unit price basis.

Complete the work, excluding plant establishment work, within 120 working days.

Complete the work, including plant establishment work, within 345 working days.

Complete the plant establishment work within 250 working days.

The estimated cost of the project is \$6,500,000.

A mandatory prebid meeting is scheduled on October 29, 2024 at 9:45 a.m. at WEBEX.

The Department will receive bids until 2:00 p.m. on the bid open date via Bid Express website. Bids received after this time will not be accepted. For more information refer to the Electronic Bidding Guide at the Office Engineer's website.

The Department will open and publicly read the bids through webcast/teleconference services immediately after the specified closing time.

For bid results go to:

<http://ppmoe.dot.ca.gov/des/oe/contractor-info.html>

Select *Electronic Bidding* under the *Bidding* tab.

District office addresses are provided in the *Standard Specifications*.

Present bidders' inquiries to the Department and view the Department's responses at:

<http://ppmoe.dot.ca.gov/des/oe/bid-inquiries.php>

Questions about alleged patent ambiguity of the plans, specifications, or estimate must be asked before bid opening. After bid opening, the Department does not consider these questions as bid protests.

Submit your bid with bidder's security equal to at least 10 percent of the bid.

Prevailing wages are required on this Contract. The Director of the California Department of Industrial Relations determines the general prevailing wage rates. Obtain the wage rates at the DIR website,

<http://www.dir.ca.gov>, or from the Department's Labor Compliance Office of the district in which the work is located.

The federal minimum wage rates for this Contract as determined by the United States Secretary of Labor are available at <https://sam.gov/content/home>.

If the minimum wage rates as determined by the United States Secretary of Labor differs from the general prevailing wage rates determined by the Director of the California Department of Industrial Relations for similar classifications of labor, the Contractor and subcontractors must not pay less than the higher wage rate. The Department does not accept lower State wage rates not specifically included in the federal minimum wage determinations. This includes helper, or other classifications based on hours of experience, or any other classification not appearing in the federal wage determinations. Where federal wage determinations do not contain the State wage rate determination otherwise available for use by the Contractor and subcontractors, the Contractor and subcontractors must not pay less than the federal minimum wage rate that most closely approximates the duties of the employees in question.

The Department has made available Notices of Suspension and Proposed Debarment from the Federal Highway Administration. For a copy of the notices, go to http://www.dot.ca.gov/hq/esc/oe/contractor_info. Additional information is provided in the Excluded Parties List System at <https://sam.gov/content/home>.

Caltrans and the Construction Industry are committed to making partnering the way we do business. For more information, go to <http://www.dot.ca.gov/hq/construc/partnering.html>.

Department of Transportation

D04YK

BID ITEM LIST

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0001	070030	LEAD COMPLIANCE PLAN	LS	LUMP SUM
0002	090100	TIME-RELATED OVERHEAD (WDAY)	WDAY	120
0003	090214	SAFETY QUALITY CONTROL MANAGER	LS	LUMP SUM
0004	120090	CONSTRUCTION AREA SIGNS	LS	LUMP SUM
0005	120100	TRAFFIC CONTROL SYSTEM	LS	LUMP SUM
0006	120120	TYPE III BARRICADE	EA	74
0007	120149	TEMPORARY PAVEMENT MARKING (PAINT)	SQFT	540
0008	120159	TEMPORARY TRAFFIC STRIPE (PAINT)	LF	6,890
0009	120165	CHANNELIZER (SURFACE MOUNTED)	EA	420
0010	120300	TEMPORARY PAVEMENT MARKER	EA	52
0011	120320	TEMPORARY BARRIER SYSTEM	LF	5,940
0012	128651	PORTABLE CHANGEABLE MESSAGE SIGN (EA)	EA	9
0013	129105	TEMPORARY CRASH CUSHION TL-2	EA	26
0014	130100	JOB SITE MANAGEMENT	LS	LUMP SUM
0015	130201	WATER POLLUTION CONTROL PROGRAM	LS	LUMP SUM
0016	130620	TEMPORARY DRAINAGE INLET PROTECTION	EA	33
0017	130640	TEMPORARY FIBER ROLL	LF	4,960
0018	130900	TEMPORARY CONCRETE WASHOUT	LS	LUMP SUM
0019	141103	REMOVE YELLOW THERMOPLASTIC TRAFFIC STRIPE (HAZARDOUS WASTE)	LF	4,430
0020	141120	TREATED WOOD WASTE	LB	1,920

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0021	153248	REMOVE CONCRETE (MISCELLANEOUS) (SQFT)	SQFT	610
0022	160110	TEMPORARY HIGH-VISIBILITY FENCE	LF	4,570
0023	170103	CLEARING AND GRUBBING (LS)	LS	LUMP SUM
0024	190101	ROADWAY EXCAVATION	CY	5,690
0025	192037	(F) - STRUCTURE EXCAVATION (RETAINING WALL)	CY	633
0026	193013	(F) - STRUCTURE BACKFILL (RETAINING WALL)	CY	338
0027	193031	(F) - PERVIOUS BACKFILL MATERIAL (RETAINING WALL)	CY	23
0028	200114	ROCK BLANKET	SQFT	2,440
0029	200123	CULTIVATION	SQYD	580
0030	202006	SOIL AMENDMENT	CY	65
0031	202037	ORGANIC FERTILIZER	LB	20
0032	204035	PLANT (GROUP A)	EA	200
0033	204099	PLANT ESTABLISHMENT WORK	LS	LUMP SUM
0034	205035	WOOD MULCH	CY	65
0035	206559	CONTROL AND NEUTRAL CONDUCTORS (ARMOR-CLAD)	LS	LUMP SUM
0036	206562	1" REMOTE CONTROL VALVE	EA	4
0037	206564	1 1/2" REMOTE CONTROL VALVE	EA	1
0038	206634	2" WYE STRAINER ASSEMBLY	EA	1
0039	206751	12 STATION IRRIGATION CONTROLLER (WALL MOUNTED)	EA	1
0040	207055	(F) - 2" COPPER PIPE (SUPPLY LINE)	LF	20

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0041	208423	1" BACKFLOW PREVENTER ASSEMBLY	EA	1
0042	037631	FLOW SENSOR CABLE	LS	LUMP SUM
0043	208442	FLOW SENSOR	EA	1
0044	208448	RISER SPRINKLER ASSEMBLY	EA	200
0045	208575	2" GATE VALVE	EA	2
0046	208594	(F) - 3/4" PLASTIC PIPE (SCHEDULE 40) (SUPPLY LINE)	LF	1,107
0047	208595	(F) - 1" PLASTIC PIPE (SCHEDULE 40) (SUPPLY LINE)	LF	142
0048	208597	(F) - 1 1/2" PLASTIC PIPE (SCHEDULE 40) (SUPPLY LINE)	LF	120
0049	208598	(F) - 2" PLASTIC PIPE (SCHEDULE 40) (SUPPLY LINE)	LF	290
0050	208601	(F) - 4" PLASTIC PIPE (SCHEDULE 40) (SUPPLY LINE)	LF	200
0051	208640	PRESSURE REGULATING VALVE	EA	1
0052	208649	QUICK COUPLING VALVE	EA	3
0053	208683	BALL VALVE	EA	1
0054	208738	8" CORRUGATED HIGH DENSITY POLYETHYLENE PIPE CONDUIT	LF	110
0055	210010	MOVE-IN/MOVE-OUT (EROSION CONTROL)	EA	5
0056	210110	IMPORTED TOPSOIL (CY)	CY	480
0057	210300	HYDROMULCH	SQFT	105,000
0058	210350	FIBER ROLLS	LF	8,380
0059	210430	HYDROSEED	SQFT	105,000
0060	210610	COMPOST (CY)	CY	170

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0061	210630	INCORPORATE MATERIALS	SQFT	21,000
0062	250201	CLASS 2 AGGREGATE SUBBASE	CY	1,370
0063	260203	CLASS 2 AGGREGATE BASE (CY)	CY	1,670
0064	390132	HOT MIX ASPHALT (TYPE A)	TON	2,710
0065	390137	RUBBERIZED HOT MIX ASPHALT (GAP GRADED)	TON	590
0066	394073	PLACE HOT MIX ASPHALT DIKE (TYPE A)	LF	21
0067	394076	PLACE HOT MIX ASPHALT DIKE (TYPE E)	LF	20
0068	397005	TACK COAT	TON	9.6
0069	398100	REMOVE ASPHALT CONCRETE DIKE	LF	20
0070	398200	COLD PLANE ASPHALT CONCRETE PAVEMENT	SQYD	1,500
0071	398300	REMOVE BASE AND SURFACING	CY	530
0072	401050	JOINTED PLAIN CONCRETE PAVEMENT	CY	160
0073	510060	(F) - STRUCTURAL CONCRETE, RETAINING WALL	CY	167
0074	510092	(F) - STRUCTURAL CONCRETE, HEADWALL	CY	5.4
0075	510094	(F) - STRUCTURAL CONCRETE, DRAINAGE INLET	CY	38
0076	048620	CONCRETE SURFACE TEXTURE (WOOD PLANK)	SQFT	1,109
0077	520103	(F) - BAR REINFORCING STEEL (RETAINING WALL)	LB	19,300
0078	568062	PEDESTRIAN BARRICADE	EA	1
0079	600025	REMOVE RETAINING WALL (PORTION) (LF)	LF	170
0080	610108	18" ALTERNATIVE PIPE CULVERT	LF	460

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0081	610112	24" ALTERNATIVE PIPE CULVERT	LF	80
0082	610300	(F) - CONCRETE BACKFILL (PIPE TRENCH)	CY	8.5
0083	700617	DRAINAGE INLET MARKER	EA	13
0084	710136	REMOVE PIPE (LF)	LF	280
0085	710150	REMOVE INLET	EA	4
0086	710152	REMOVE HEADWALL	EA	2
0087	710262	CAP INLET	EA	1
0088	721420	CONCRETE (DITCH LINING)	CY	9
0089	730020	MINOR CONCRETE (CURB) (CY)	CY	70
0090	730070	DETECTABLE WARNING SURFACE	SQFT	330
0091	731504	MINOR CONCRETE (CURB AND GUTTER)	CY	160
0092	731507	MINOR CONCRETE (GUTTER DEPRESSION)	CY	1.3
0093	731521	MINOR CONCRETE (SIDEWALK)	CY	200
0094	731710	REMOVE CONCRETE CURB (LF)	LF	4,340
0095	731780	REMOVE CONCRETE SIDEWALK (SQYD)	SQYD	1,350
0096	731790	REMOVE CONCRETE ISLAND (PORTIONS) (SQYD)	SQYD	260
0097	733000	PRE/POST CONSTRUCTION SURVEYS	EA	10
0098	750001	(F) - MISCELLANEOUS IRON AND STEEL	LB	3,977
0099	780256	ADJUST ACCESS BOX (UTILITY)	EA	4
0100	016142	RELOCATE BUS SHELTER AND TRASH RECEPTACLE	EA	1

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0101	780440	PREPARE AND STAIN CONCRETE	SQFT	1,160
0102	800103	TEMPORARY FENCE (TYPE CL-6)	LF	2,810
0103	800360	CHAIN LINK FENCE (TYPE CL-6)	LF	170
0104	802501	4' CHAIN LINK GATE (TYPE CL-6)	EA	1
0105	803050	REMOVE CHAIN LINK FENCE	LF	130
0106	810120	REMOVE PAVEMENT MARKER	EA	360
0107	810230	PAVEMENT MARKER (RETROREFLECTIVE)	EA	500
0108	820270	REMOVE ROADSIDE SIGN (WOOD POST)	EA	35
0109	820280	REMOVE ROADSIDE SIGN (METAL POST)	EA	10
0110	820300	REMOVE ROADSIDE SIGN (STRAP AND SADDLE BRACKET METHOD)	EA	3
0111	820310	REMOVE ROADSIDE SIGN PANEL	EA	6
0112	820530	RESET ROADSIDE SIGN	EA	1
0113	820750	FURNISH SINGLE SHEET ALUMINUM SIGN (0.063"-UNFRAMED)	SQFT	540
0114	820760	FURNISH SINGLE SHEET ALUMINUM SIGN (0.080"-UNFRAMED)	SQFT	99
0115	820780	FURNISH SINGLE SHEET ALUMINUM SIGN (0.063"-FRAMED)	SQFT	44
0116	820790	FURNISH SINGLE SHEET ALUMINUM SIGN (0.080"-FRAMED)	SQFT	159
0117	820840	ROADSIDE SIGN - ONE POST	EA	46
0118	820850	ROADSIDE SIGN - TWO POST	EA	7
0119	820900	INSTALL ROADSIDE SIGN PANEL ON EXISTING POST	EA	5
0120	015334	GREEN THERMOPLASTIC PAVEMENT MARKING	SQFT	520

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0121	840617	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 6-1)	LF	90
0122	840619	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 12-3)	LF	110
0123	840621	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 17-7)	LF	940
0124	014765	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 9-3)	LF	400
0125	846007	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY)	LF	9,490
0126	846008	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 8-4)	LF	350
0127	846009	8" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY)	LF	1,380
0128	846010	8" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 12-3)	LF	140
0129	846012	THERMOPLASTIC CROSSWALK AND PAVEMENT MARKING (ENHANCED WET NIGHT VISIBILITY)	SQFT	3,030
0130	846013	12" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY)	LF	480
0131	846020	REMOVE PAINTED TRAFFIC STRIPE	LF	759
0132	846025	REMOVE PAINTED PAVEMENT MARKING	SQFT	12
0133	846030	REMOVE THERMOPLASTIC TRAFFIC STRIPE	LF	5,710
0134	846035	REMOVE THERMOPLASTIC PAVEMENT MARKING	SQFT	1,300
0135	847218	6" TRAFFIC STRIPE TAPE WITH CONTRAST (WARRANTY)	LF	560
0136	847221	6" TRAFFIC STRIPE TAPE WITH CONTRAST (WARRANTY) (BROKEN 17-7)	LF	280
0137	870136	ELECTRIC SERVICE FOR IRRIGATION	LS	LUMP SUM
0138	872131	MODIFYING LIGHTING SYSTEMS	LS	LUMP SUM
0139	872133	MODIFYING SIGNAL AND LIGHTING SYSTEMS	LS	LUMP SUM
0140	995100	WATER METER CHARGES	LS	LUMP SUM

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0141	995200	IRRIGATION WATER SERVICE CHARGES	LS	LUMP SUM
0142	999990	MOBILIZATION	LS	LUMP SUM

SPECIAL PROVISIONS

ORGANIZATION

Special provisions are under headings that correspond with the main-section headings of the *Standard Specifications*. A main-section heading is a heading shown in the table of contents of the *Standard Specifications*.

Each special provision begins with a revision clause that describes or introduces a revision to the *Standard Specifications*.

Any paragraph added or deleted by a revision clause does not change the paragraph numbering of the *Standard Specifications* for any other reference to a paragraph of the *Standard Specifications*.

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DIVISION I GENERAL PROVISIONS

1 GENERAL

Add to section 1-1.01:

Bid Items and Applicable Sections

Item code	Item description	Applicable section
090214	SAFETY QUALITY CONTROL MANAGER	5
995100	WATER METER CHARGES	5
995200	IRRIGATION WATER SERVICE CHARGES	5
999990	MOBILIZATION	9
037631	FLOW SENSOR CABLE	20
048620	CONCRETE SURFACE TEXTURE (WOOD PLANK)	51
016142	RELOCATE BUS SHELTER AND TRASH RECEPTACLE	78
014765	6" THERMOPLASTIC TRAFFIC STRIPE (ENHANCED WET NIGHT VISIBILITY) (BROKEN 9-3)	84
015334	GREEN THERMOPLASTIC PAVEMENT MARKING	84

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2 BIDDING

Replace section 2-1.04 with:

2-1.04 MANDATORY VIRTUAL PREBID OUTREACH MEETING

The Department will conduct a mandatory virtual prebid meeting to provide small businesses, including DVBs and DBs, the opportunity to meet and interact with prospective bidders in an effort to increase their participation in the performance of contracts.

The virtual meeting will be a video conference meeting using Cisco Webex.

Each bidder must attend the virtual meeting. The bidder's representative must be a company officer, project superintendent, or project estimator. For a joint venture, one of the parties must attend the virtual meeting.

The Department does not accept a bid from a bidder who did not attend the virtual meeting.

Bidders and small businesses must register a minimum of 1 hour before the meeting with:

DUTY SENIOR, DISTRICT 4 OFFICE

111 GRAND AVENUE

OAKLAND, CA 94612

TELEPHONE: (510) 286-5209

E-MAIL: D4.Construction.Duty.Senior@dot.ca.gov

An electronic sign-in sheet will be used to identify the attendees. Each bidder must include the name and title of the company representative attending the virtual meeting.

The sign-in sheets, with the names of all companies in attendance at the virtual meeting, will be made available at the website shown on the *Notice to Bidders* for bidder inquiries.

The successful bidder is required to report each small business hired to work on this Contract as a result of the meeting.

Add between the 1st and 2nd paragraphs of section 2-1.06B:

The Department makes the following supplemental project information available:

Supplemental Project Information

Means	Description
Included in the <i>Information Handout</i>	1. Stormwater Information Handout 2. Water Efficient Landscape Worksheet 3. Warranty Bond (TOTE-1 Form) 4. Geotechnical Design Report 5. Final Report-Utilization of Small Business Enterprise (DOT OCR-0003 Form)
Available as specified in the <i>Standard Specifications</i>	Cross Sections
Included with the project plans	Logs of test borings

Replace section 2-1.11 with:

2-1.11 IN-USE OFF-ROAD DIESEL-FUELED VEHICLE LIST

Section 2-1.11 applies to non-informal-bid contracts.

Complete and submit the In-Use Off-Road Diesel-Fueled Vehicle List form under section 2-1.33.

On the In-Use Off-Road Diesel-Fueled Vehicle List form, list each fleet used by you or your subcontractor to perform work and is subject to 13 CCR § 2449 et seq. Submit a copy of a valid Certificate of Reported Compliance (13 CCR § 2449, subdivision (n)) for each fleet listed on the form within 10 days of bid opening. Failure to list a fleet used by you or your subcontractor to perform work on the In-Use Off-Road Diesel-Fueled Vehicle List form may result in a nonresponsive bid. Failure to submit the Certificate of Reported Compliance for a fleet listed on the In-Use Off-Road Diesel-Fueled Vehicle List form may result in a nonresponsive bid.

Replace section 2-1.33B with:

2-1.33B Bid Form Submittal Schedules

2-1.33B(1) General

The *Bid* book includes forms specific to the Contract. The deadlines for the submittal of the forms vary depending on the requirements of each Contract. Determine the requirements of the Contract and submit the forms based on the applicable schedule specified in section 2-1.33B.

Bid forms and information on the form that are due after the time of bid may be submitted at the time of bid.

2-1.33B(2) Federal-Aid Contracts

2-1.33B(2)(a) General

Section 2-1.33B(2) applies to a federal-aid contract.

2-1.33B(2)(b) Contracts with a DBE Goal

2-1.33B(2)(b)(i) General

Section 2-1.33B(2)(b) applies if a DBE goal is shown on the *Notice to Bidders*.

2-1.33B(2)(b)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Federal-Aid Contract with a DBE Goal	
Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number
Copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid except for the public works contractor registration number
Copy of the Subcontractor List as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Small Business Status	Time of bid
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
DBE Commitment	No later than 4 p.m. on the 5th day after bid opening ^b
DBE Confirmation	No later than 4 p.m. on the 5th day after bid opening ^b
DBE Good Faith Efforts Documentation	No later than 4 p.m. on the 5th day after bid opening ^b

^aSubmit only if you choose the option.

^bIf the last day for submitting the bid form falls on a Saturday or holiday, it may be submitted on the next business day with the same effect as if it had been submitted on the day specified.

2-1.33B(2)(b)(iii) Reserved

2-1.33B(2)(c) Contracts without a DBE Goal

2-1.33B(2)(c)(i) General

Section 2-1.33B(2)(c) applies if a DBE goal is not shown on the *Notice to Bidders*.

2-1.33B(2)(c)(ii) Bid Form Schedule

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Federal-Aid Contract without a DBE Goal

Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number
Copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid except for the public works contractor registration number
Copy of the Subcontractor List as submitted at the time of bid with the public works contractor registration numbers	10 days after bid opening
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Small Business Status	Time of bid
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid

^aSubmit only if you choose the option.

2-1.33B(2)(c)(iii) Reserved

2-1.33B(2)(d)–2-1.33B(2)(h) Reserved

2-1.33B(3) Non-Federal-Aid Contracts

2-1.33B(3)(a) General

Section 2-1.33B(3) applies to non-federal-aid contracts.

2-1.33B(3)(b) Contracts with a DVBE Goal

2-1.33B(3)(b)(i) General

Section 2-1.33B(3)(b) applies if a DVBE goal is shown on the *Notice to Bidders*.

2-1.33B(3)(b)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Non-Federal-Aid Contract with a DVBE Goal

Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number for a joint-venture contract
For a joint-venture contract, copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
Certified DVBE Summary	No later than 4 p.m. on the 4th business day after bid opening
California Company Preference	Time of bid
Request for Small Business Preference or Non–Small Business Preference ^a	Time of bid
Certified Small Business Listing for the Non–Small Business Preference ^a	No later than 4 p.m. on the 2nd business day after bid opening

^aSubmit only if you choose the option or preference.

2-1.33B(3)(b)(iii) Reserved

2-1.33B(3)(c) Contracts without a DVBE Goal

2-1.33B(3)(c)(i) General

Section 2-1.33B(3)(c) applies if a DVBE goal is not shown on the *Notice to Bidders*.

2-1.33B(3)(c)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Non-Federal-Aid Contract without a DVBE Goal	
Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number for a joint-venture contract
For a joint-venture contract, copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
California Company Preference	Time of bid
Certified DVBE Summary ^b	No later than 4 p.m. on the 4th business day after bid opening
Request for Small Business Preference or Non-Small Business Preference ^a	Time of bid
Certified Small Business Listing for the Non-Small Business Preference ^a	No later than 4 p.m. on the 2nd business day after bid opening

^aSubmit only if you choose the option or preference.

^bSubmit only if you obtain DVBE participation or you are the apparent low bidder, 2nd low bidder, or 3rd low bidder and you choose to receive the specified incentive.

2-1.33B(3)(c)(iii) Reserved

2-1.33B(3)(d)–2-1.33B(3)(h) Reserved

2-1.33B(4)–2-1.33B(9) Reserved

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5 CONTROL OF WORK

Replace the 2nd and 3rd paragraphs of section 5-1.09A with:

Professionally facilitated project partnering is required.

Replace section 5-1.13D with:

5-1.13D Non-Small Business Preference

5-1.13D(1) General

Section 5-1.13D applies to a non-federal-aid contract when you have obtained a non-small business subcontractor preference under section 2-1.18.

Use each SBE as shown on the Certified Small Business Listing for the Non-Small Business Preference form unless you receive authorization for a substitution under section 5-1.13D(3).

Maintain records of subcontracts and agreements made with each SBE. Include in the records:

1. SBE subcontract or agreement
2. Name and business address of each business
3. Proof of each payment made to an SBE

Upon reasonable notice and during normal business hours, permit access to the SBE premises for the purpose of:

1. Interviewing employees.
2. Inspecting and copying books, records, accounts, and other material that may be relevant to a matter under investigation.

Upon work completion, complete a Final Report – Utilization of Small Business Enterprise form and submit within 10 days of Contract acceptance. The Department withholds the greater of 10 percent of the SBE work amount or \$10,000 until the form is submitted. The Department releases the withhold upon submission of the completed form. If additional payments are made to an SBE after submittal of the completed form, submit an updated form to reflect such payments.

5-1.13D(2) Commercially Useful Function

SBEs must perform a commercially useful function under 2 CA Code Regs § 1896.15 when performing tasks listed on the Certified Small Business Listing for the Non-Small Business Preference form and as a Department authorized substitute SBE. The SBE's value of work will only count toward the non-small business preference if the SBE performs a commercially useful function under 2 CA Code Regs § 1896.15.

Provide written notification to the Engineer at least 15 days in advance of each SBE's initial performance of work on the Contract. Include the SBE's name, contract work to be performed, and the location, date, and time of where the SBE's work will take place.

Monitor each SBE's performance of a commercially useful function throughout their duration of work on the Contract. Maintain supporting documentation of commercially useful function related records. Provide copies of the following to the Department within 5 business days of a specific SBE request:

1. Subcontract or agreement with the SBE
2. Purchase orders
3. Bills of lading
4. Invoices
5. Proof of payment

You and your SBEs must submit any commercially useful function related records and documents within 5 business days of Department request such as:

1. Proof of ownership or lease and rental agreements for equipment
2. Tax records
3. Employee rosters
4. Certified payroll records
5. Inventory rosters

Notify the Engineer immediately if you believe an SBE may not be performing a commercially useful function.

Failure to submit requested records and documents, or failure of an SBE to perform a commercially useful function, will be a violation of these specifications. Such violations will result in referral to the Department of General Services, and you may be subject to sanctions under Govt Code § 14842 et seq.

5-1.13D(3) Substitutions

Substitution of an SBE may be requested for the reasons listed in Pub Cont Code § 4107. The Department will consider your SBE substitution request under 2 CA Code Regs § 1896.10. The substitute must be another SBE.

The requirement that an SBE be certified by the bid opening date does not apply to SBE substitutions after contract award.

Include in your substitution request:

1. Copy of the written notice issued to the SBE with proof of delivery
2. Copy of the SBE response to the notice
3. Your explanation of the reason for the substitution request
4. Names and certification numbers of the listed SBE and the proposed substitute SBE
5. Description of the work to be performed including dollar amount or percentage of overall contract the proposed substitute SBE will perform
6. If an SBE substitution cannot occur, written justification and the steps that were taken to try to acquire a new SBE and how that non-small business preference portion of the Contract will be fulfilled

Upon receipt of the substitution request, the listed SBE will be notified by the Department of the substitution request. The listed SBE will have 5 business days to respond. Where the listed SBE objects to the substitution, the Department will schedule a hearing.

If the Department authorizes your SBE substitution request, it will do so in writing.

Work performed by a firm other than the listed SBE or authorized substitute will be a violation of these specifications. Such violations will result in notification of the Department of General Services, and you may be subject to sanctions under Govt Code § 14842 et seq.

Replace section 5-1.13E with:

5-1.13E Prompt Payment

Section 5-1.13E applies to all contracts.

Pay your subcontractors within 7 days of receipt of each progress payment under Pub Cont Code §§ 10262 and 10262.5. Pay duly authorized motor carriers of property in dump trucks for transportation charges under Bus & Prof Code § 7108.6. Pay other entities, such as material suppliers, within 30 days of receipt of each progress payment.

Each month, after the 15th and prior to 20th, submit the following payment information through the Department's prompt payment monitoring system at <https://caltrans.dbesystem.com>:

1. Subcontractor's or entity's business name
2. Description of work performed
 - 2.1. Bid item numbers or change order numbers
 - 2.2. Written narrative of work performed
3. Value of work performed
4. Amount paid to subcontractor or entity
5. Withhold amount, if applicable
6. Explanation of withhold reasoning, if applicable

Your subcontractors and other entities may validate payments received using the prompt payment monitoring system.

If a subcontractor's or other entity's work is in dispute, provide a written withhold notification to the subcontractor or entity and the Engineer no later than 7 days after receipt of the corresponding progress payment that includes the following:

1. Value of the disputed work

2. Amount of the withhold being taken
3. Bid item numbers or change order numbers associated with the disputed work
4. Explanation of the deficiencies of the disputed work and how the corresponding value was calculated
5. Corrective actions to be taken for release of withheld amount

The Department may request additional documentation from you to evaluate whether you applied the withhold in good faith. Submit requested documents within 10 days of receipt of request.

The Department may withhold the same amount of your withhold from a future progress pay estimate if the Department determines any of the following has occurred:

1. Withhold was not applied in good faith
2. Requested additional withhold documentation records were not provided
3. Payment information was not submitted through the prompt payment monitoring system
4. Required withhold notification was not provided

The Department may also apply a 2 percent penalty on the withhold amount for every month payment is not made.

Replace section 5-1.14 with:

5-1.14 SAFETY QUALITY CONTROL MANAGER

5-1.14A General

Allocate a full-time, on-site safety quality control manager (SQCM) to project safety, for the duration of this Contract. The SQCM must be available after hours as needed. The SQCM is not required to be on-site during a plant establishment period or when no work is being performed during project temporary suspensions.

Develop a written site-specific safety program (SSSP) that incorporates known hazards associated with the project.

5-1.14A(1) Submittals

Submit a resume of the SQCM before starting work on the project, including education, heavy construction safety experience, and completed safety training.

Submit a copy of the SQCM's certificates for Cal/OSHA 30-hour construction training course and traffic control supervisor training as an informational submittal. The certificate must include:

1. Names of Department authorized entity providing certification
2. Name of the individual receiving certification
3. Date when the certification was provided
4. Expiration date

Submit a copy of the SSSP as informational submittals:

1. At least 15 days before starting work
2. At least 48 hours before starting work whenever a revision is made to the SSSP

Submit daily safety status reports within two business days of report date, including completed construction site safety inspection checklists as informational submittals.

5-1.14A(2) Quality Assurance

5-1.14A(2)(a) General

Not Used

5-1.14A(2)(b) Certifications

The SQCM must have completed training and received a certificate of completion for:

1. Cal/OSHA 30-hour construction training course

2. Traffic control supervisor certification. Department authorized traffic control supervisor training providers list is available at:

<https://dot.ca.gov/programs/construction/safety-traffic/safety-training-courses>

The SQCM must have one of the following certifications from the Board of Certified Safety Professionals:

1. Certified Safety Professional
2. Construction Health and Safety Technician
3. Occupational Hygiene and Safety Technician

5-1.14A(2)(c) Qualifications

The SQCM must be a competent and qualified person, as defined by Cal/OSHA, and must provide safety oversight on the project. The SQCM must have:

1. Four-year college degree
2. At least 5 years of experience in heavy construction safety
3. Knowledge in the area of safety procedure for all types of work being performed on this project
4. Knowledge of Caltrans specifications and plans, Cal/OSHA policy and procedures, and California Code of Regulations Title 8 safety requirements

The SQCM must have completed safety training and maintain valid certifications for:

1. First aid
2. Cardiopulmonary resuscitation (CPR)
3. Confined space
4. Fall protection

5-1.14B Materials

Not Used

5-1.14C Construction

The SQCM must be available by:

1. Cellular telephone
2. Two-way radio
3. Mobile internet access

The SQCM must:

1. Develop the SSSP
2. Confirm each worker's compliance with the Cal/OSHA drug testing requirements
3. Conduct the on-site new project orientation for each worker assigned to the project
4. Review the construction work plans for each subcontractor before starting work
5. Conduct or attend pre-planning sessions for high hazard work such as excavations, demolition, confined space entry, falsework, crane hoisting, high-risk utilities such as, high pressure pipelines with fuel or natural gas, and tunneling or boring
6. Conduct weekly job site safety meetings
7. Review work schedules including subcontractors', to ensure adequacy of job hazard analyses and address all safety concerns and issues
8. Ensure SDSs are available on-site for all materials and have a hazard communication program for informing workers of material hazards
9. Verify compliance with hazardous waste requirements under section 14-11, including spill prevention and control measures
10. Conduct daily safety inspection of the job site for hazards and compliance with safety requirements in the specifications and 8 CA Code of Regs
11. Document safety inspections at every location where construction activity is active using a construction site safety inspection checklist that includes at a minimum:
 - 11.1. Date
 - 11.2. Name of inspector

- 11.3. Inspection location
- 11.4. Construction activity
- 11.5. Name of foreperson
- 11.6. Safety topics that include:
 - 11.6.1. General
 - 11.6.2. Housekeeping and sanitation
 - 11.6.3. Fire prevention
 - 11.6.4. Personal protective equipment
 - 11.6.5. Electrical safety
 - 11.6.6. Hand and power tools
 - 11.6.7. Falsework safety
 - 11.6.8. Scaffolding safety
 - 11.6.9. Ladder safety
 - 11.6.10. Excavations and trenches
 - 11.6.11. Heavy equipment
 - 11.6.12. Cranes
 - 11.6.13. Concrete construction
 - 11.6.14. Steel erection
 - 11.6.15. Welding and cutting
 - 11.6.16. HMA or concrete paving
 - 11.6.17. Tunneling
 - 11.6.18. Confined space
 - 11.6.19. Fall protection
- 12. Ensure that the lane closure traffic control daily report is completed by the traffic control technicians and traffic control daily review report is completed by the traffic control supervisor when a traffic control supervisor is required
- 13. Review the lane closure traffic control daily reports and traffic control daily review reports for any work zone related safety issues
- 14. Revise SSSP to include procedures to mitigate hazardous situations identified in construction site safety inspections, lane closure traffic control daily reports, and traffic control daily review reports
- 15. Investigate any near misses or reported incidents in a timely manner and take appropriate action to prevent similar incidents in the future
- 16. Report all incidents to the Engineer immediately, including the investigation of any close calls
- 17. Prepare a daily safety status report that includes:
 - 17.1. Daily construction work activities, including temporary traffic control
 - 17.2. Safety related field activities
 - 17.3. Identified safety deficiencies and corrective actions taken
 - 17.4. Meetings, conversations, and other communications related to safety issues
 - 17.5. Any reported near misses or incidents
- 18. Submit daily safety status reports to the Contractor's on-site superintendent for submittal to the Engineer

When monitoring work zone, take immediate corrective action and notify the Engineer if an imminent danger is identified.

If the Engineer determines the SQCM is not meeting the intended duties and responsibilities, you must replace the SQCM within 30 days of written notice from the Engineer. If you need to replace the SQCM, notify the Engineer and provide replacement SQCM resume and required certifications at least 15 days before the replacement.

5-1.14D Payment

Not Used

Replace section 5-1.20E with:

5-1.20E Water Meter Charges

Section 5-1.20E applies if a bid item for water meter charges is shown on the Bid Item List.

The local water authority is Marin Municipal Water District.

The local water authority will install the water meters.

The charges are as shown in the following table:

Water Meter Charges		
Meter size	Quantity	Charge per meter (\$)
1"	1	35,000

The charges by the local water authority include:

1. Furnishing and installing each water meter
2. Connecting to the local water authority's main water line, including any required hot tap or tee
3. Furnishing and installing an extension pipe from the main water line to the water meter
4. Sterilizing the extension pipe

Make arrangements and pay the charges for installation of the water meters.

If the charge is changed at the time of installation, the Department adjusts the lump sum price based on the difference between the specified charges and the changed charges.

Replace section 5-1.20F with:

5-1.20F Irrigation Water Service Charges

Contact the local water authority to arrange a start date for water service and pay for the service until Contract acceptance.

The local water authority is Marin Municipal Water District.

The charge for water service from the water authority is \$307/bi-monthly plus the charges shown in the following table:

Water Service Charges		
Meter size	Amount of water (gallons)	Charge (\$)
1.5"	8,228	119.46

Submit a copy of each monthly water bill.

Notify the Engineer upon paying your final bill.

If the local water authority's charges are changed, the Department adjusts the lump sum price based on the difference between the specified charges and the changed charges.

Replace section 5-1.24B with:

5-1.24B Department Construction Surveys for Automated Machine Guidance

The Department sets control points to a minimum of 0.07 foot local horizontal accuracy and 3rd order vertical accuracy standards.

For slope stakes and rough grade stakes, the Department sets 6 survey control points or 2 per mile, whichever is greater.

The Department sets slope stakes and rough grade stakes at:

1. Conform stations
2. Beginning and end of each alignment
3. Midpoint or every 200 feet, whichever results in a greater number of stakes, on a curve
4. Every 500 feet on tangents

The Department sets final grade stakes under Chapter 12, "Construction Surveys," section 12.5-6 of the Department's *Surveys Manual*.

At your request, the Department sets survey control points under section 12.1-6, "Automated Machine Guidance." When control stakes are requested, final grade stakes are set at:

1. Conform stations
2. Beginning and end of each alignment
3. Midpoint or every 100 feet, whichever results in the greater number of stakes, on a curve with a radius of 1,200 feet or less
4. Midpoint or every 200 feet, whichever results in the greater number of stakes, on a curve with a radius of more than 1,200 feet
5. Every 200 feet on a tangent

At your request and under Chapter 12 of the Department's *Surveys Manual*, the Department provides (1) staking for intersections, clearing, fencing, drainage, curbs, structures, abutment fill, wall, and miscellaneous areas and (2) additional survey control or staking for earthwork in areas where global navigation satellite system coverage is inadequate for automated machine guidance.

Replace section 5-1.25 with:

5-1.25 AUTOMATED MACHINE GUIDANCE

5-1.25A General

You may use automated machine guidance (AMG) if the AMG meets or exceeds the staking tolerances described in section 12.5, "Typical Department-Furnished Construction Stakes," of the Department's *Surveys Manual*.

You are responsible for determining whether the work and site conditions are practical for AMG use.

Furnish a global navigation satellite system (GNSS) rover compatible with your GNSS base station or the GNSS correction service you subscribe to. The Department returns the GNSS rover upon work completion. This is change order work.

At the preconstruction conference, be prepared to discuss survey control points, site and equipment calibration, inspection methods, conflict resolution, and staking.

5-1.25B Definitions

automated machine guidance: Technology that uses positioning devices, singly or in combination, such as global navigation satellite systems (GNSS), total stations, or rotating laser levels, to determine and control the real-time position of construction equipment using onboard computer equipment.

California Coordinate System of 1983 (CCS83): CCS83 as defined in Pub Res Code § 8801.

digital construction model: Three-dimensional model used by the Contractor's AMG equipment.

digital design model: Three-dimensional model consisting of roadway design alignments, profiles, and cross sections representing the finished grade.

digital terrain model: Three-dimensional model representing the original ground before job site activities start.

global navigation satellite system: Satellite system used to pinpoint the geographic location of a user's receiver anywhere in the world. Two GNSS systems are in operation: the US GPS and the Russian Federation's GLONASS. Each of the GNSS systems uses a constellation of orbiting satellites working in conjunction with a network of ground stations.

GNSS base station: Single ground-based system consisting of a GNSS receiver, antenna, and telemetry equipment that provides differential GNSS correction signals to other GNSS receivers or rovers. Multiple base stations can be combined into a GNSS network.

GNSS correction service subscription: Subscription service to receive differential GNSS correction signals for higher accuracy GNSS positioning without the need of a GNSS base station. Signals are normally received via cellular wireless data services.

GNSS rover: Portable GNSS antenna, receiver, rod, and data collector with telemetry equipment for real-time point measurements.

grid: Cartesian coordinate system of Northing (y) and Easting (x) coordinates using CCS83.

robotic total station: Survey instrument capable of tracking an optical target and providing real-time coordinates of the target to the equipment operator and AMG equipment. A robotic total station unit can provide AMG if site conditions do not allow GNSS receivers to be used and if a higher accuracy is required than the GNSS provides.

site calibration or localization: Process that establishes the relationship between the observed control point coordinates and the site coordinate system, which is usually grid. The term applies to both GNSS and robotic total station equipment.

5-1.25C Electronic Files

Electronic design files include:

1. Digital terrain model in 3-D DGN or LandXML format
2. Roadway design alignments and profiles in LandXML format
3. Cross sections in 2-D DGN and PDF
4. Digital design model in LandXML format

The Department makes electronic design files available as supplemental project information.

You must create the digital construction model (DCM).

Convert the electronic design files to a format compatible with your AMG system. Manipulation of the electronic design files is at your own risk.

Submit copies of the digital construction model files and any updates to them in LandXML format.

Digital design model information may not exist for contour grading and some drainage areas. The Department places stakes for these areas.

The Department provides you with updated electronic data whenever the Engineer determines a plan change materially affects the finished grade. For minor grade changes, the Department places stakes and marks.

5-1.25D Quality Control Plan

Submit an AMG QC plan at least 15 days before starting work requiring AMG. The plan must include the following information:

1. Contract number
2. Name and contact information of the AMG QC technician
3. Limits of the area for which the AMG will be used
4. Scope of work to be completed using AMG for the following work categories:
 - 4.1. Clearing and grubbing
 - 4.2. Earthwork
 - 4.3. Trench excavation
 - 4.4. Rough grading
 - 4.5. Subgrade
 - 4.6. Subbase
 - 4.7. Base
 - 4.8. Curb and gutter
 - 4.9. Cold planning or milling existing pavement
 - 4.10. Paving
 - 4.11. Intelligent compaction
 - 4.12. Concrete barrier
 - 4.13. Finishing roadway
5. Project control plan sheet detailing control points covering the job site
6. List of GNSS equipment, including:
 - 6.1. Type

- 6.2. Manufacturer
- 6.3. Model
- 6.4. Software version
- 7. Description of GNSS site calibration or localization checking, including:
 - 7.1. List of equipment requiring calibration or localization checking
 - 7.2. Site calibration or localization procedures
 - 7.3. Frequency of calibration or localization
 - 7.4. Format for recording calibrations or localizations, including:
 - 7.4.1. Date
 - 7.4.2. Locations where calibration or localization was performed
 - 7.4.3. GNSS equipment manufacturer and model
 - 7.4.4. Range of required tolerance
 - 7.4.5. Name and signature of the person performing calibration or localization
 - 7.5. Reporting time for submitting records of calibration or localization
- 8. Description of daily GNSS equipment or robotic total station equipment check-testing procedures, including the format for recording daily check testing
- 9. List of AMG onboard computer equipment, including:
 - 9.1. Type
 - 9.2. Manufacturer
 - 9.3. Software version
 - 9.4. List of AMG-controlled equipment, including:
 - 9.4.1. Type, such as loader or grader
 - 9.4.2. Manufacturer
 - 9.4.3. Model
- 10. Procedures for AMG-controlled equipment calibration, including:
 - 10.1. Description of equipment calibration procedures
 - 10.2. Frequency of calibration
 - 10.3. Format for recording calibration information
- 11. Electronic data file control, including:
 - 11.1. Name and contact information of the person responsible for the electronic files
 - 11.2. DCM file-naming convention
 - 11.3. Description of the process that will be used to upload the DCM to the AMG equipment
 - 11.4. Description of the process that will be used whenever updated DCM files are required to be uploaded to the AMG equipment

If QC procedures or personnel change, submit a QC plan supplement describing the change.

5-1.25E Quality Control Technician

During AMG activities, provide a QC technician to be responsible for:

- 1. GNSS site calibration or localization and upload to all GNSS receivers
- 2. Maintenance of GNSS and AMG equipment
- 3. Documentation of the calibration or localization and maintenance of GNSS equipment
- 4. Daily calibration and documentation of AMG equipment
- 5. Daily setup and takedown of GNSS and robotic total station components

5-1.25F Just-in-Time Training

Provide at least 8 hours of just-in-time (JIT) training on the GNSS rover for up to 3 Department employees. Provide training materials and equipment.

The JIT training must cover the following topics:

- 1. Background information for the GNSS to be used
- 2. Setup and calibration checks for:
 - 2.1. GNSS receiver
 - 2.2. GNSS base station
 - 2.3. GNSS rovers
 - 2.4. Machinery
- 3. Operation of the GNSS rover, including:
 - 3.1. Setup data collection

- 3.2. Settings for alignments and profiles
- 3.3. Onboard display options
4. Demonstration of grade checking using the rover

The training is change order work.

5-1.25G Construction

5-1.25G(1) General

If you find a discrepancy in any survey control point, survey stake, or in the electronic data provided, immediately, submit an RFI.

5-1.25G(2) GNSS Site Calibration or Localization

Perform GNSS site calibration or localization to the survey control points at least 5 business days before starting work requiring AMG.

Check each survey control point for accuracy. Submit the GNSS site calibration or localization results within 1 business day of the calibration or localization testing. Allow 3 business days for the review of the results

5-1.25G(3) GNSS Check Testing

Before starting daily work requiring AMG, conduct check testing for the proper setup of the GNSS or robotic total station equipment. Ensure the GNSS or robotic total station equipment achieves accuracies within:

1. 0.10 foot in both horizontal and vertical directions for rough grading
2. 0.05 foot in horizontal directions and 0.02 foot in vertical directions for final grades

Before starting daily production, conduct check testing of the AMG equipment and the GNSS rovers.

Within 1 business day after check testing, submit the check-testing results as informational submittals.

5-1.25G(4) Grade Verification

If requested, provide a GNSS rover and personnel to operate it for the Engineer's use in verifying grades. This is change order work.

Replace section 5-1.26 with:

5-1.26 GRADE QUALITY CONTROL

Use a global navigation satellite system (GNSS) rover, robotic total station equipment, or a level to check the grades at the frequencies shown in the following table:

Grade Checking Requirements

Type of work	Area or distance represented by the grade checking	Frequency (number of grade points)
Earthwork for cut and fill slopes ≤ 15 feet	200 feet	2
Earthwork for cut and fill slopes > 15 feet	1,000 sq yd	1
Rough grading	1,000 sq yd	1
Trenching	100 feet	6
Subgrade	1 mi	30
Subbase layer	1 mi	50
Base layer	1 mi	100
Curb and gutter	100 feet	6
Concrete barrier	100 feet	5
Finishing roadway	1,000 sq yd	2

Increase the frequency of grade checking of a roadway:

1. Wherever its curve radius is 500 feet or less
2. In areas of a superelevation transition
3. At intersections

Notify the Engineer when an area is ready for line and grade inspection. Submit the grade checking results on a Grade Checking Report form as an informational submittal.

Add to the end of section 5-1.32:

Personal vehicles of your employees must not be parked on the traveled way or shoulders, including sections closed to traffic.

The utilities shown in the following table may interfere with the work and must be exposed and protected in place. Make arrangements with the utility owner (1) to conduct or witness all exposures or (2) to request temporary deactivation of the utility.

Utilities to Be Exposed and/or Protected in Place During Construction

Utility	Location	Contact
Pacific Gas & Electric 4" Gas Line	At intersection of Bayshore Freeway/Civic Center Drive and	Edgardo Ochoa, Gas Engineering
Pacific Gas & Electric 4" Electric Line	Manuel Freitas Parkway 41.92' Rt C1 "Sta 73+25 to 73+40 41' Lt "PSR" Sta 11+46 to 15+29	(925) 322-3315 or via email EEO1@pge.com

Within the tolerance zone of the PG&E gas line, use only hand tools to excavate and expose the line. Do not use power-operated or power-driven excavation or boring equipment within 24 inches horizontally of the exterior surface of the subsurface facility unless otherwise authorized by the engineer or the utility operator.

For locations within 18 inches vertically above the gas line or 24 inches horizontally from the gas line (1) use hand-held or hand-guided compacting equipment and (2) the work must be coordinated with PG&E Gas Engineering.

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6 CONTROL OF MATERIALS

Add to the beginning of section 6-1.02:

The Department furnishes you with:

1. Loop detector sensor units
2. Controller assemblies for traffic signals, ramp metering and pedestrian hybrid beacons
3. Modems
4. Electronics assembly for battery backup systems as follows:
 - 4.1. Inverter/charger unit

The Department furnishes you with completely wired controller cabinets with auxiliary equipment but without controller unit at 30 Rickard Street, San Francisco CA 94134. At least 48 hours before you pick up the materials, inform the Engineer of what you will pick up and when you will pick it up.

You must furnish replacement plants. The Department does not pay you for the replacement plants.

Replace section 6-1.03B with:

6-1.03B Submittals

6-1.03B(1) General

Not Used

6-1.03B(2) Work Plan

For local material, such as rock, gravel, earth, structure backfill, pervious backfill, imported borrow, and culvert bedding, obtained from a (1) noncommercial source, or (2) source not regulated under California jurisdiction, submit a local material plan for each material at least 60 days before placing the material. The local material plan must include:

1. Certification signed by you and an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State stating:

I am aware local material from a noncommercial source or a source not regulated under CA jurisdiction must be sampled and analyzed for pH and lead and may require sampling and analysis under section 6-1.03B(3) for other constituents of concern based on the land use history. I am aware that local material sources must not contain ADL at concentrations greater than 80 mg/kg total lead or equal to or greater than 5 mg/L soluble lead as determined by the Waste Extraction Test (WET) Procedures, 22 CA Code of Regs § 66261.24(a)(2) App II. I am aware that a maximum quantity of material may be excavated at the site based on the minimum number of samples taken before excavating at the site under section 6-1.03B(3).

2. Land use history of the local material location and surrounding property
3. Sampling protocol
4. Number of samples per volume of local material
5. QA and QC requirements and procedures
6. Qualifications of sampling personnel
7. Stockpile history
8. Name and address of the analytical laboratory that will perform the chemical analyses
9. Analyses that will be performed for lead and pH
10. Other analyses that will be performed for possible hazardous constituents based on:
 - 10.1. Source property history
 - 10.2. Land use adjacent to source property
 - 10.3. Constituents of concern in the ground water basin where the job site is located

The plan must be sealed and signed by an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State.

If the plan requires revisions, the Engineer provides comments. Submit a revised plan within 7 days of receiving comments. Allow 7 days for the review.

6-1.03B(3) Analytical Test Results

At least 15 days before placing local material, submit analytical test results for each local material obtained from a noncommercial source or a source not regulated under CA jurisdiction. The analytical test results must include:

1. Certification signed by an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State stating:

The analytical testing described in the local material plan has been performed. I performed a statistical analysis of the test results of the analytical testing described in the local material plan using the US EPA's ProUCL software with the applicable 95 percent upper confidence limit. I certify that the material from the local material source is suitable for unrestricted use at the job site and the material has met the following criteria:

1. Has a pH above 5.0.

2. Does not contain soluble lead in concentrations equal to or greater than 5 mg/L as determined by the Waste Extraction Test (WET) Procedures, 22 CA Code of Regs § 66261.24(a)(2) App II.
3. Does not contain lead in concentrations above 80 mg/kg total lead.
4. Is not contaminated with the other constituents of concern identified in the local material plan in average concentration (95 percent upper confidence limit) in excess of these constituents' respective San Francisco Bay RWQCB commercial/industrial environmental screening levels ESLs, except for arsenic.
5. Does not exceed the maximum allowed concentration limit table listed in Section 6-1.03B(4).

2. Chain of custody of samples.
3. Analytical results no older than 1 year.
4. Statistical analysis of the data using US EPA's ProUCL software with a 95 percent upper confidence limit.
5. Comparison of sample results and 95 percent upper confidence limits to hazardous waste concentration thresholds and the applicable San Francisco Bay RWQCB environmental screening levels (ESLs) given in direct exposure human health risk levels (Table S-1), commercial/industrial: Shallow soil exposure, under Summary of Soil ESLs tables (2019 Rev 2). The Summary of Soil ESLs tables (2019 Rev 2) can be obtained by sending an email to ESLs.ESLs@waterboards.ca.gov with "Request for ESL Documents" in the subject line.

6-1.03B(4) Sample and Analysis

Sample and analyze local material from a (1) noncommercial source or (2) a source not regulated under CA jurisdiction:

1. Before bringing the local material to the job site
2. As described in the local material plan
3. Under US EPA Test Methods for Evaluating Solid Waste, Physical/Chemical Methods (SW-846)

The sample collection must be designed to generate a data set representative of the entire volume of proposed local material.

Before excavating at (1) a noncommercial material source or (2) a source not regulated under CA jurisdiction, collect the minimum number of samples, and perform the minimum number of analytical tests for the corresponding maximum volume of local material as shown in the following table:

Minimum Number of Samples and Analytical Tests for Local Material

Maximum volume of imported borrow (cu yd)	Minimum number of samples and analytical tests
< 5,000	8
5,000–10,000	12 for the first 5,000 cu yd plus 1 for each additional 1,000 cu yd or portion thereof
10,000–20,000	17 for the first 10,000 cu yd plus 1 for each additional 2,500 cu yd or portion thereof
20,000–40,000	21 for the first 20,000 cu yd plus 1 for each additional 5,000 cu yd or portion thereof
40,000–80,000	25 for the first 40,000 cu yd plus 1 for each additional 10,000 cu yd or portion thereof
> 80,000	29 for the first 80,000 cu yd plus 1 for each additional 20,000 cu yd or portion thereof

Do not collect composite samples or mix individual samples to form a composite sample.

Statistically analyze the samples' laboratory results using the US EPA's ProUCL software to define 95 percent upper confidence limit for the various contaminants of concern. All chemical analysis must be performed by a laboratory certified by the SWRCB's Environmental Laboratory Accreditation Program (ELAP).

The analytical test results must demonstrate that the local material:

1. Is not a hazardous waste
2. Has a pH above 5.0
3. Has an average total lead concentration, based upon the 95 percent upper confidence limit, at or below 80 mg/kg
4. Is not contaminated with local material plan-identified constituents of concern at average concentrations (95 percent upper confidence limits) in excess of their respective commercial/industrial San Francisco Bay RWQCB environmental screening levels ESLs, except for arsenic.
5. Does not contain any of the following compounds, chemicals, or elements at an estimated average concentration (95 percent upper confidence limit) above the maximum allowed concentration defined in the following table:

Compound/Chemical	Maximum allowed concentration (mg/kg)
Arsenic	11
Barium	1500
Benzene	1
Beryllium	10
Cadmium	10
Chromium (total)	1000
Cobalt	100
Diesel	150
Ethylbenzene	10
Gasoline	500
Mercury	10
Motor oil	500
Nickel	150
Selenium	10
Toluene	10
Trichloroethene	1
Vanadium	200
Xylenes	10
Zinc	600

6-1.03C Local Material Management

Do not place local material until authorized.

If the Engineer determines the appearance, odor, or texture of any delivered local material suggests possible contamination, sample and analyze the material. The sampling and analysis is change order work unless (1) hazardous waste is discovered or (2) the analytical test results indicate the material does not comply with section 6-1.03B(3).

Dispose of noncompliant local material at an appropriately permitted CA Class I, CA Class II or CA Class III facility. You are the generator of noncompliant local materials.

Replace section 6-1.04 with:

6-1.04 BUY AMERICA

6-1.04A General

Buy America requirements do not apply to the following:

1. Tools and construction equipment used in performing the work
2. Temporary work that is not incorporated into the finished project

6-1.04B Crumb Rubber (Pub Res Code § 42703(d))

Furnish crumb rubber with a certificate of compliance. Crumb rubber must be:

1. Produced in the United States
2. Derived from waste tires taken from vehicles owned and operated in the United States

6-1.04C Steel and Iron Materials

Steel and iron materials must be melted and manufactured in the United States except:

1. Foreign pig iron and processed, pelletized, and reduced iron ore may be used in the domestic production of the steel and iron materials
2. If the total combined cost of the materials produced outside the United States does not exceed the greater of 0.1 percent of the total bid or \$2,500, the material may be used if authorized

Furnish steel and iron materials to be incorporated into the work with certificates of compliance and certified mill test reports. Mill test reports must indicate where the steel and iron were melted and manufactured.

All melting and manufacturing processes for these materials, including an application of a coating, must occur in the United States. Coating includes all processes that protect or enhance the value of the material to which the coating is applied.

6-1.04D Manufactured Products

Iron and steel used in precast concrete manufactured products must meet the requirements of section 6-1.04C regardless of the amount used.

Iron and steel used in other manufactured products must meet the requirements of section 6-1.04C if the weight of steel and iron components constitute 90 percent or more of the total weight of the manufactured product.

6-1.04E Construction Materials

The following construction materials must be produced in the United States under standards in 2 CFR 184.6:

1. Non-ferrous metals
2. Plastic and polymer-based products such as:
 - 2.1. Polyvinylchloride
 - 2.2. Composite building materials
3. Glass
4. Fiber optic cable including drop cable
5. Optical fiber
6. Lumber
7. Engineered wood
8. Drywall

All manufacturing processes for these materials as defined in 2 CFR 184.6 must occur in the United States.

Furnish construction materials to be incorporated into the work with certificates of compliance with each project delivery. Manufacturer's certificate of compliance must identify where the construction material was manufactured and attest specifically to compliance with its 2 CFR 184.6 standard.

Minor additions of articles, materials, supplies, or binding agents to these construction materials do not change the categorization of the construction material.

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7 LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

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Replace the row for *Schedule* in the table in the 2nd paragraph of section 8-1.03 with:

Replace section 8-1.04C with:

Section 8-1.04B does not apply.

Do not start job site activities until the Department authorizes or accepts your submittal for:

- You may enter the job site only to measure controlling field dimensions and locate utilities.

1. Notice of Materials To Be Used form.
2. Written statement from the vendor that the order for the sign panels has been received and accepted by the vendor. The statement must show the dates that the materials will be shipped.
3. Written statement from the vendor that the order for electrical material has been received and accepted by the vendor. The statement must show the dates that the materials will be shipped.
4. Written statement from the vendor that the order for structural steel has been received and accepted by the vendor. The statement must show the dates that the materials will be shipped.

1. Obtain specified authorization or acceptance for each submittal before the 55th day
2. Receive authorization to start

Replace the 4th and 5th paragraphs of section 8-1.05 with:

The weekly statement of working days report shows:

- Contract No. 04-0K8004

2. Time adjustments
3. Calculations of work completion dates, including working days remaining
4. Controlling activities

Delete the 6th paragraph of section 8-1.05.

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9 PAYMENT

Replace the 1st paragraph of section 9-1.11D with:

For progress payments, the total work completed for the TRO bid item is the number of working days shown for the pay period on the weekly statement of working days report.

[illegible]

DIVISION II GENERAL CONSTRUCTION

10 GENERAL

Add to the end of section 10-1.02C(2):

Protect any irrigation component to be relocated before performing any other construction activity in the area.

AA

12 TEMPORARY TRAFFIC CONTROL

Add to section 12-3.11B(5)(b):

Provide two C48A(CA) construction project funding identification signs.

Legend for the type of project must read as follows:

HIGHWAY IMPROVEMENT

Legend for the year of completion on project funding sign must read as follows:

YEAR OF COMPLETION 2024

Add the following funding partner agency pictographs:

Partner funding agency	Pictograph details
City of San Rafael	City of San Rafael logo

Replace section 12-3.20 with:

12-3.20 TEMPORARY BARRIER SYSTEMS

12-3.20A General

12-3.20A(1) Summary

Section 12-3.20 includes specifications for placing, maintaining, repairing, and removing temporary barrier systems.

Temporary barrier system consists of:

1. New or undamaged used interconnected barrier segments
2. Segment connection hardware
3. Stakes and anchor bolts

12-3.20A(2) Definitions

clear area width: Minimum width throughout the length of the barrier system that must be maintained clear of obstructions, objects, and work resources during non-working hours. The width is measured perpendicular from the non-traffic side toe.

set back distance: Space measured between the closest toe of temporary barrier and the edge of traveled way for each direction of traffic.

12-3.20A(3) Submittals

Submit as informational submittal for each type of temporary barrier system:

1. Certificate of compliance.
2. Manufacturer's installation instructions except for temporary concrete barriers with loop and pin and temporary concrete barriers with cross bolt.
3. Manufacturer's QC test results and daily production log, through the Data Interchange for Materials Engineering (DIME) website. QC test results must include the concrete mix design number, barrier stamped ID, and must be submitted within 3 business days of QC test completion.

Submit test reports for cross bolts that certify compliance with the applicable ASTM requirements. The test reports must be from a laboratory that is accredited to International Standards Organization/International Electrotechnical Commission 17025 by the American Association for Laboratory Accreditation (A2LA) or the ANSI-ASQ National Accreditation Board.

Submit a signed manufacturer's replacement evaluation report within 10 days of damage to a temporary steel barrier system.

12-3.20A(4) Quality Assurance

12-3.20A(4)(a) General

Temporary barrier systems must comply with MASH Test Level 3 except for Type K temporary railing.

Except for temporary concrete barriers with loop and pin and temporary concrete barriers with cross bolt, temporary barrier systems must:

1. Be on the Authorized Materials List for highway safety features
2. Comply with the manufacturer's drawings shown on the Department's Division of Safety Programs website and the manufacturer's installation instructions

If a discrepancy exists, governing ranking in descending order is:

1. These specifications
2. Manufacturer's drawings
3. Manufacturer's installation instructions

QC sampling, testing, and inspection personnel must have an ACI Concrete Field-Testing Technician, Grade I certification.

Temporary concrete barrier segments must:

1. Comply with the requirements for tier 3 precast concrete in section 90-4
2. Be fabricated at a plant on the Authorized Facility Audit List

Concrete must be sampled and tested at the minimum frequencies shown in the following table.

Concrete QC Tests		
Quality characteristic	Test method	Minimum testing frequency
Compressive strength	ASTM C172/C172M, ASTM C31/C31M, and ASTM C39/C39M	Once per 300 cu yd of concrete cast, or every day of casting, whichever is more frequent
Slump	ASTM C143/C143M	
Temperature at time of mixing	ASTM C1064/C1064M	
Density	ASTM C138	Once per 600 cu yd of concrete cast or every 7 days of batching, whichever is more frequent
Air content	ASTM C231/C231M or ASTM C173/C173M	If concrete is air entrained, once for each set of cylinders, and when conditions warrant

A daily production log of PC activities must be maintained under section 90-4.01C(4).

12-3.20A(4)(b) Quality Control

Replace damaged temporary concrete barrier segments with exposed reinforcing steel or concrete spalls 1-1/2 inches in depth and 4 inches in width or greater.

Replace damaged temporary steel barrier segments with permanent bends, tearing, or buckling as described in the signed manufacturer's replacement evaluation report.

Realign temporary barrier system within 2 days of impact or displacement when displaced more than 3 inches except when the temporary barrier system is displaced into a traveled lane realign immediately.

12-3.20B Materials

12-3.20B(1) General

Temporary barrier segment must:

1. Be a minimum 31-1/2 inches in height
2. Have at least two lifting holes
3. Be designed to be used with temporary traffic screen when required

Temporary barrier segment may have your name or logo on each barrier segment. The name or logo must be no more than 4 inches in height and must be located no more than 12 inches above the bottom of the barrier segment.

12-3.20B(2) Temporary Concrete Barriers

12-3.20B(2)(a) General

Temporary concrete barrier segment must:

1. Be precast concrete with a minimum 4,000-psi compressive strength.
2. Have reinforcement steel that complies with section 52.
3. Have a finished surface that complies with section 51-1.03F(2).
4. Include the manufacturer's name, lot number, and month and year of manufacture stamped on the top of each barrier segment except for Type K temporary railing. The stamped information must be:
 - 4.1. No more than 6 inches in height.
 - 4.2. From 3/16 to 1/4 inch in depth.

- 4.3. Centered on the top width of the barrier segment.
5. Use one of the following segment connections:
 - 5.1. Loop and pin
 - 5.2. "J" hook
 - 5.3. Cross bolt
6. Comply with the tolerances shown in the following table:

Precast Barrier Tolerances

Quality Characteristic	Tolerance(±)
Length (in)	1
Insert Placement (in)	1/2
Horizontal Alignment (in)	1/8 per 10 feet of length
Deviation of Ends (in):	
Horizontal Skew	1/4
Vertical Batter	1/8 per foot of depth

Reinforcement steel must:

1. Comply with ASTM A615 or ASTM A706, Grade 60
2. Be galvanized under section 52-3, when required

Combinations of reinforcing steel and welded wire reinforcement are allowed. Welded wire reinforcement must comply with ASTM A1064.

Stake must:

1. Comply with ASTM A36/A36M-14 or ASTM A529-14, Grade 50
2. Be 1-1/2-inch-diameter-by-48-inch-long
3. Have a plate 1/2-by-3-1/2-by-3-1/2-inch welded 2 inches down from the upper end using a 1/4-inch fillet weld under AWS D1.1 or D1.4

Anchor bolt must:

1. Be a threaded rod, 1-1/8-inch-diameter-by-10-1/2-inch-long
2. Comply with ASTM A307
3. Include a nut complying with ASTM A563
4. Include a plate washer:
 - 4.1. 1/2-by-3-1/2-by-3-1/2-inch with a 1-1/4-inch diameter hole in the center
 - 4.2. Complying with ASTM A36/A36M
 - 4.3. Galvanized post fabrication under section 75-1.02B

Epoxy adhesive must have a bond strength of minimum 1,650 psi, except for temporary concrete barrier with "J" Hook.

12-3.20B(2)(b) Temporary Concrete Barriers with Cross Bolt

Cross bolt hardware includes:

1. Cross bolt
2. Nut complying with ASTM A563
3. Hardened washer complying with ASTM F436, Type 1
4. Plate washer complying with ASTM A36/A36M and galvanized post fabrication under section 75-1.02B

Cross bolt must:

1. Be a 7/8-inch bolt or threaded rod and comply with one of the following:
 - 1.1. HS threaded rod ASTM A193, Grade B7
 - 1.2. HS threaded rod ASTM A449, Type 1
 - 1.3. HS nonheaded anchor bolt ASTM F1554, Grade 105, Class 2A
2. Have a permanent grade symbol and manufacturer's identifier

12-3.20B(2)(c) Temporary Concrete Barriers with Loop and Pin

12-3.20B(2)(c)(i) General

Steel bar loop must comply with ASTM A36/A36M.

Connecting bolt must comply with ASTM A307, be 1-1/4-inch in diameter, and a minimum 26-inch length. A round bar of the same diameter and length may be substituted for the connecting bolt. The round bar must:

1. Comply with ASTM A36/A36M
2. Have a 3-inch-diameter, 3/8-inch-thick plate welded on the upper end using a 3/16-inch fillet weld

12-3.20B(2)(c)(ii) Type K Temporary Railings

Anchor bolt must:

1. Be a threaded rod, 1-inch-diameter-by-15-1/2-inch-long
2. Comply with ASTM A307
3. Include a nut complying with ASTM A563
4. Include a plate washer:
 - 4.1. 3/8-by-2-1/2-by-3-inch with a 1-1/8-inch diameter hole in the center
 - 4.2. Complying with ASTM A36/A36M
 - 4.3. Galvanized post fabrication under section 75-1.02B

12-3.20B(2)(d) Temporary Concrete Barriers with "J" Hook

"J" hook must:

1. Comply with ASTM A36/A36M
2. Be 3/8-inch-thick steel plate
3. Be a minimum 18 inches in height

Anchor hardware must include:

1. Anchor bolt insert 1-inch diameter, 6-inch long
2. Hex head bolt 1-inch diameter with a minimum length of 11 inches plus thickness of asphalt overlay
3. Plate washer 3/8-inch by 3-inch by 3-inch
4. Retainer ring

12-3.20B(3) Temporary Steel Barriers

Temporary steel barrier segment must:

1. Be galvanized steel.
2. Have a joint connection.
3. Include permanent identification information with no more than 6 inches in height and 12 inches in length and centered on the top width of the segment. The identification information must include:
 - 3.1. Manufacturer's name.
 - 3.2. Serial number.
 - 3.3. Lot number.
 - 3.4. Month and year of manufacture.

19-foot temporary steel barrier segment must be filled to a depth of 11-13/16 inches with concrete ballast.

12-3.20C Construction

12-3.20C(1) General

Clean temporary barrier segments at time of installation and at least every 6 months thereafter.

Install temporary barrier systems based on the requirements shown in the following table:

Minimum Clear Area Width

Barriers	Configuration	Height differentials 3 feet or less (ft)	Height differentials greater than 3 ft up to 8 feet (ft)	Edge of deck or height differentials greater than 8 feet (ft)	Fixed objects, falsework members, or temporary supports ^a (ft)
10-foot, 20-foot & 30-foot temporary concrete barriers with cross bolt	Freestanding	1	2	5	5
12.5-foot temporary concrete barriers with "J" hook	Freestanding	3	4	8	7
	3 stakes per segment traffic side	1	1	2	3
	2 anchor bolts per segment traffic side	1	1	2	3
20-foot temporary concrete barriers with "J" hook	Freestanding	3	4	8	7
	4 stakes per segment traffic side	1	1	2	3
	3 anchor bolts per segment traffic side	1	1	2	3
50-foot temporary steel barriers	Staked or anchored at both ends only	6	7	9	10
	Staked or anchored every 250 feet	5	6	8	9
	Staked or anchored every 33 feet	1	1	3	4
19-foot temporary steel barriers	Freestanding	4	5	7	8
20-foot Type K temporary railings	Freestanding	2	3	8	7
	2 stakes or 2 anchor bolts per segment traffic side	1	1	3	4
	4 stakes or 4 anchor bolts per segment	N/A	N/A	3	3

^aThe minimum clear area width to a falsework or temporary support footing can be 2 feet less than the clear area width shown. Measure clear area width to the footing edge closest to traffic.

Stake temporary barrier systems when placed on an asphalt concrete surface.

Anchor temporary barrier systems when placed on a concrete surface. For bridge decks, confirm the anchor will not penetrate closer than 1-1/2 inches from the bottom of the deck before placement. When temporary barrier is not shown, request the Engineer to verify the bridge deck thickness.

For installations on concrete surfaces, drill holes and bond threaded rods or dowels under section 51-1.03E(5). Do not drill the top of supporting beams or girders, bridge expansion joints, or drains.

Install stakes and anchor bolts so the heads do not project above the top of the temporary barrier pocket profile.

In addition to the temporary barrier minimum length required, for the approach zone before the protected area, place a minimum:

1. 60 feet temporary barrier on facilities with a posted speed of 45 mph or less
2. 100 feet temporary barrier on facilities with a posted speed greater than 45 mph

Offset the approach end of temporary barrier systems a minimum of 15 feet from the edge of an open traffic lane, use the offset rate shown in the following table:

Temporary Barrier System Offset Rate	
Posted speed (mph)	Rate ^a
0 to 45	10:1
46 to 60	15:1
61 to 70	20:1

^aRate is longitudinally to transversely with respect to the edge of the traveled way

If a 15-foot minimum offset cannot be achieved, offset temporary barrier systems the maximum distance available and install an authorized temporary crash cushion system at each barrier approach end.

Install a reflector on the top or face of barrier segments placed within 10 feet of a traffic lane. Space reflectors at approximately 20-foot intervals. Apply adhesive for mounting the reflector under the reflector manufacturer's instructions.

Install a Type P marker panel complying with section 82 at:

1. Each end of a temporary barrier system placed adjacent to a two-lane, two-way highway
2. The end facing traffic for a temporary barrier system installed adjacent to a one-way roadbed
3. The end of the skew nearest the traveled way when a temporary barrier system is placed on a skew

Maintain a minimum height of 31-1/2 inches above surface for temporary barrier systems. For paving activities adjacent to temporary barriers, do not pave within 2 feet of the barrier segments unless authorized. For paving under the temporary barrier, remove and reset the barrier.

Remove temporary barrier systems when no longer required for the work. Remove stakes and anchor bolts so that minimal damage is done to surface.

After removing the temporary barrier systems:

1. Restore the area to its previous condition or construct it to its planned condition if temporary excavation or embankment was used to accommodate the temporary barrier.
2. Remove all threaded rods or dowels to a depth of at least 1 inch below the top of a concrete surface. Fill the resulting holes with mortar under section 51-1 except cure the mortar by the water method or by the curing compound method using curing compound no. 6.
3. Repair a damaged asphalt surface by providing a clean, smooth edge around the damaged area. Repair any heaving caused by stake removal to provide a uniform surface. Remove loose debris and use compressed air to clean out the stake hole. Comply with manufacturer's requirements except fill the stake hole with grout to existing pavement elevation under section 51-1.

If the Engineer orders a lateral move of a temporary barrier system and repositioning is not shown, the lateral move is change order work except for work area access, clear area width compliance, or because of your means and methods to perform the work.

12-3.20C(2) Temporary Concrete Barriers

12-3.20C(2)(a) General

Before placing temporary barrier systems on the job site and after each described relocation, paint the exposed surfaces of the segments with white paint complying with specifications for acrylic emulsion paint for exterior masonry.

Place and maintain the abutting ends of segments in alignment without substantial offset from each other.

Install temporary barrier systems with the last segment extending a minimum of 60 feet past the length of the protected area.

Stake or anchor a minimum 20 feet of barrier at each end of temporary barrier systems. For:

1. Temporary concrete barriers with "J" hook, place a minimum of 6 stakes or anchors at each end, 3 on each side.
2. Temporary concrete barriers with cross bolt, place a minimum of 6 stakes or anchors at each end, 3 on each side.
3. Type K temporary railings, place 4 stakes or anchors at each end, 2 on each side.

Maintain a minimum 1-foot set back distance for temporary barrier systems placed between two-way traffic.

12-3.20C(2)(b) Temporary Concrete Barriers with Cross Bolt

Install a minimum 210 feet of temporary concrete barrier with cross bolt.

Place temporary barrier systems on a concrete or asphalt concrete surface.

Do not stake or anchor temporary barrier systems, except for 20 feet at each end.

Intermix segments of different lengths within a temporary barrier system when necessary.

For temporary barrier systems placed on a curved layout, maintain the minimum curve radius shown in the following table:

Minimum Curve Radius	
Segment length (ft)	Curve radius (ft)
10	125
20	265
30	400

12-3.20C(2)(c) Temporary Concrete Barriers with Loop and Pin

12-3.20C(2)(c)(i) General

Not Used

12-3.20C(2)(c)(ii) Type K Temporary Railings

Do not install Type K temporary railings on projects advertised after December 31, 2026.

Install a minimum 160 feet of Type K temporary railing.

Excavate and backfill under section 19-3.

Do not compact earth fill placed behind Type K temporary railings in a curved layout.

Place temporary barrier systems on a firm, stable surface. Grade the area to provide a uniform bearing surface throughout the entire length of the system.

Anchor or stake the first and last segment and every other segment with four stakes as shown, when placed between two-way traffic.

12-3.20C(2)(d) Temporary Concrete Barriers with "J" Hook

Install a minimum 200 feet of temporary concrete barrier with "J" hook.

Place temporary barrier systems on a concrete or asphalt concrete surface. The asphalt concrete surface must have a minimum 2 inches of asphalt concrete over 6 inches of compacted subbase.

When temporary barrier systems are placed between two-way traffic, install on each side as shown:

1. 2 anchors or stakes for 12.5-foot segments
2. 3 anchors or stakes for 20-foot segments

12-3.20C(3) Temporary Steel Barriers

12-3.20C(3)(a) General

Install temporary barrier systems under manufacturer's instructions.

Maintain a minimum 2-foot set back distance on both sides of temporary barrier systems placed between two-way traffic.

12-3.20C(3)(b) 50-Foot Temporary Steel Barriers

Use 50-foot temporary steel barriers with or without rubber pads.

Install a minimum 250 feet of 50-foot temporary steel barrier. The last segment must extend a minimum 25 feet past the length of the protected area.

Place temporary barrier systems on a concrete or asphalt concrete surface. Do not place systems on a dirt surface.

Anchor or stake the first and last segment of temporary barrier systems.

Maintain a minimum radius of 800 feet for segments placed on a curved layout. For tighter curves down to a 250-foot radius, contact the manufacturer before installation and provide manufacturer's written recommendation for the installation.

12-3.20C(3)(c) 19-Foot Temporary Steel Barriers

Install a minimum 323 feet of 19-foot temporary steel barrier.

Stake the first and last segment of temporary barrier systems.

Maintain a minimum radius of 262 feet for segments placed on a curved layout.

12-3.20D Payment

The payment quantity for temporary barrier systems is the length measured along the top of the barrier segments.

Replace section 12-3.22 with:

12-3.22 TEMPORARY CRASH CUSHIONS

12-3.22A General

12-3.22A(1) Summary

Section 12-3.22 includes specifications for installing, repairing, replacing, maintaining, and removing temporary crash cushions.

12-3.22A(2) Definitions

Not Used

12-3.22A(3) Submittals

At least 10 days before installation, submit as informational submittal for each temporary crash cushion model:

1. Certificate of compliance
2. Two copies of the following:
 - 2.1. Manufacturer's installation and maintenance manual
 - 2.2. Department approved manufacturer's drawings from the Department's Division of Safety Programs website
3. Record of training provided by manufacturer for each person installing the temporary crash cushion

12-3.22A(4) Quality Assurance

12-3.22A(4)(a) General

Not Used

12-3.22A(4)(b) Quality Control

Temporary crash cushion must be installed under the manufacturer's instructions by personnel trained by the manufacturer.

Keep a copy of the manufacturer's drawings, and installation and maintenance manual for each temporary crash cushion model at the job site during installation.

12-3.22B Materials

12-3.22B(1) General

Temporary crash cushion must:

1. Be on the Authorized Materials List for highway safety features
2. Comply with MASH:
 - 2.1. Test Level 3 (TL-3) for a posted speed limit 45 mph or greater
 - 2.2. Test Level 2 (TL-2) for a posted speed limit less than 45 mph

Water-filled temporary crash cushion must:

1. Include all components required for attachment to temporary barrier or protected obstacle
2. Comply with the manufacturer's drawings shown on the Department's Division of Safety Programs website

Each sand-filled temporary crash cushion module must be:

1. Colored standard yellow with a black lid
2. Free from structural flaws and objectionable surface defects

Sand for filling module must:

1. Be commercial quality, washed concrete sand
2. Contain no more than 5 percent water under California Test 226
3. Be clean when placed in the module

12-3.22B(2) Temporary Crash Cushions TL-3

Temporary crash cushion TL-3 must:

1. Comply with MASH TL-3
2. Be one of the following:
 - 2.1. Water-filled temporary crash cushion gating, non-redirective system
 - 2.2. Sand-filled temporary crash cushion module array

12-3.22B(3) Temporary Crash Cushions TL-2

Temporary crash cushion TL-2 must:

1. Comply with MASH TL-2
2. Be one of the following:
 - 2.1. Water-filled temporary crash cushion gating, non-redirective system
 - 2.2. Sand-filled temporary crash cushion module array

12-3.22C Construction

12-3.22C(1) General

When activities expose traffic to fixed obstacles, protect traffic from the obstacles with temporary crash cushions.

You may use NCHRP Report 350 compliant temporary crash cushions before December 31, 2026.

Install temporary crash cushions under the manufacturer's instructions before:

1. Starting activities requiring the crash cushions.
2. Opening to traffic lanes adjacent to the protected obstacles.

Temporary crash cushions must not impede the flow of traffic or encroach on the traveled way.

Attach a Type R or Type P marker panel to the front of temporary crash cushions if the closest point to the traveled way is within 12 feet of the traveled way. Fasten marker panels firmly to crash cushions with commercial quality hardware or by other authorized methods.

Maintain temporary crash cushions in place at each location, including when work is not in progress. Keep the area behind the temporary crash cushions clear of obstructions.

Repair damaged temporary crash cushions immediately. Remove and replace temporary crash cushions damaged beyond repair. Repair and replacement of temporary crash cushions damaged by traffic is change order work.

12-3.22C(2) Water-Filled Temporary Crash Cushions

Place temporary crash cushions and a minimum 20 feet of temporary barrier at a parallel 2-foot offset from edge of traveled way.

Install temporary crash cushions adjacent to a:

1. Barrier at an offset distance 1 to 2 feet or 7 feet or greater from the barrier
2. Dike or curb at an offset distance 0 to 4 feet or 7-1/2 feet or greater from the dike or curb

12-3.22C(3) Sand-Filled Temporary Crash Cushions

Do not use sand-filled temporary crash cushions for permanent installations.

Use the same type of sand-filled modules for each array. Fill each sand-filled module with sand under the manufacturer's instructions.

Securely fasten the top edge of a seal to the wall of the sand-filled module with a continuous strip of heavy-duty tape, when a seal is required.

You may place sand-filled temporary crash cushion modules on movable pallets or frames complying with the dimensions shown. The pallets or frames must provide a full-bearing base beneath the modules. Do not move the modules and supporting pallets or frames by sliding or skidding along the pavement or bridge deck.

Attach a Type R marker panel such that the top of the panel is 1 inch below the module lid. Attach a Type P marker panel such that the bottom of the panel rests upon the roadway surface or pallet surface when used.

You may remove sand-filled modules during the work shift for access to the work area if the exposed fixed obstacle is 15 feet or more from the nearest lane carrying traffic. Reset the modules before the end of the work shift.

A lateral move of a temporary crash cushion module is change order work if ordered and the repositioning is not shown.

Remove sand-filled temporary crash cushion modules, including sand, pallets or frames, and marker panels, at Contract acceptance.

12-3.22D Payment

The payment quantity does not include:

1. Temporary crash cushions placed for public safety
2. Modules placed in excess of the number described

Add to the beginning of section 12-3.32C:

Place PCMSs at the locations shown and in advance of the 1st warning sign for each:

1. Stationary lane closure
2. Off-ramp closure
3. On-ramp closure

Add between the 9th and 10th paragraphs of section 12-3.32C:

Start displaying the message on the sign 5 minutes before closing the lane or shoulder or when directed by the Engineer.

Replace the table in the definition of designated holidays in section 12-4.02A(2) with:

Designated Holidays	
Holiday	Date observed
New Year's Day	January 1st
Washington's Birthday	3rd Monday in February
Memorial Day	Last Monday in May
Independence Day	July 4th
Labor Day	1st Monday in September
Veterans Day	November 11th
Thanksgiving Day	4th Thursday in November
Christmas Day	December 25th

Add to section 12-4.02A(2):

special days: Third Monday in January and Halloween.

Add between the 1st and 2nd paragraphs of section 12-4.02A(3)(c):

Submit a contingency plan for each of the following activities:

1. Rapid-set concrete activities
2. HMA paving
3. Striping
4. Cold-planing asphalt concrete for depths of 2 inches or greater

Add to the end of section 12-4.02C(1):

Keep the full width of the traveled way open to traffic when no active construction activities are occurring in the traveled way or within 6 feet of the traveled way.

Keep the full width of the ramp traveled way open for use by traffic on designated holidays and special days.

For each 10-minute interval or fraction thereof past the time specified to open the closure, the amount for liquidated damages per interval shown in the table below is deducted. Liquidated damages are limited to 5 percent of the total bid per occurrence. Liquidated damages are not assessed if the Engineer orders the closure to remain in place beyond the scheduled pickup time.

Type of facility	Route	Direction or segment	Period	Liquidated damages/interval
Ramp	MRN-101	NB PM 13.759; NB PM 13.519; NB PM 13.630; SB PM 13.665; SB PM 13.883;	1st half hour 2nd half hour 2nd hour and beyond	\$1,000/10 minutes \$1,000/10 minutes \$1,000/10 minutes
City Street	Frontage Rds	NB & SB	1st half hour 2nd half hour 2nd hour and beyond	\$1,000/10 minutes \$1,000/10 minutes \$1,000/10 minutes
City Street	Manuel T Freitas Pkwy	EB & WB	1st half hour 2nd half hour 2nd hour and beyond	\$1,000/10 minutes \$1,000/10 minutes \$1,000/10 minutes

Replace section 12-4.02C(3)(f) with:

12-4.02C(3)(f) Closure Restrictions for Designated Holidays and Special Days

Closure restrictions for designated holidays and special days are shown in the following table:

Lane Closure Restrictions For Designated Holidays And Special Days											
Thu	Fri	Sat	Sun	Mon	Tues	Wed	Thu	Fri	Sat	Sun	Mon
x	H xx	xx	xx								
	SD xx										
x	xx	H xx	xx								
		SD xx									
	x	xx	H xx	xx							
			SD xx								
	x	xx	xx	H xx	xxx						
	x	xx	xx	SD xx	xxx						
				x	H xx						
				x	SD xx						
					x	H xx					
						SD xx					
						x	H xx	xx	xx	xx	
							SD xx				
Legend:											
	Refer to lane requirement charts.										
x	The full width of the traveled way must be open for use by traffic after 4:00AM.										
xx	The full width of the traveled way must be open for use by traffic.										
xxx	The full width of the traveled way must be open for use by traffic until 10:00PM.										
H	Designated holiday										
SD	Special day										

Replace section 12-4.02C(3)(j) with:

12-4.02C(3)(j) Complete Ramp Closure Hour Charts and Ramp Lane Requirement Charts

Comply with the requirements for the complete ramp closures shown in the following charts:

Chart No. J1 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/SB							Post Mile:13.665											
Closure limits: SB Loop On-ramp from EB Manuel T Freitas Parkway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C	C	C															C	C	C	C	
Fri	C	C	C	C	C	C															C	C	C	C	
Sat	C	C	C	C	C	C	C	C													C	C	C	C	
Sun	C	C	C	C	C	C	C	C													C	C	C	C	
Legend:																									
C	Ramp may be closed completely.																								
REMARKS: Detour traffic via Detour Plan No. 2 on MI-2.																									

Chart No. J2 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/SB							Post Mile:13.665											
Closure limits: SB Loop On-ramp from EB Manuel T Freitas Parkway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C																					
Fri																								C	C
Sat	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C
Sun	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C
Legend: C Ramp may be closed completely.																									
REMARKS: Detour traffic via Detour Plan No. 2 on MI-2. This Lane Closure Chart may be used up to two times. Ramp should be open on Monday at 4 am.																									

Chart No. J3 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/SB							Post Mile:13.883											
Closure limits: SB Off-ramp to Manuel T Freitas Parkway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C	C	C																C	C	C	C
Fri	C	C	C	C	C	C																	C	C	C
Sat	C	C	C	C	C	C	C	C														C	C	C	C
Sun	C	C	C	C	C	C	C	C														C	C	C	C
Legend:																									
C	Ramp may be closed completely.																								
REMARKS: Detour traffic via Detour Plan No. 1 on MI-1.																									

Chart No. J4 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/NB							Post Mile:13.519											
Closure limits: NB Off-ramp to WB Manuel T Freitas Parkway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C	C																		C	C	
Fri	C	C	C	C	C																		C	C	
Sat	C	C	C	C	C	C	C																C	C	
Sun	C	C	C	C	C	C	C	C															C	C	
Legend:																									
C	Ramp may be closed completely.																								
REMARKS: Detour traffic via Detour Plan No. 9 on MI-9.																									

Chart No. J5 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/NB							Post Mile:13.630											
Closure limits: NB On-ramp from EB Manuel T Freitas Parkway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C	C																			C	C
Fri	C	C	C	C	C																			C	C
Sat	C	C	C	C	C	C	C																	C	C
Sun	C	C	C	C	C	C	C	C																C	C
Legend:																									
C	Ramp may be closed completely.																								
REMARKS: Detour traffic via Detour Plan No. 8 on MI-8.																									

Chart No. J6 Complete Ramp Closure Hours																									
County: MRN							Route/Direction: US 101/NB							Post Mile:13.759											
Closure limits: NB On-ramp from Redwood Highway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon–Thu	C	C	C	C	C																			C	C
Fri	C	C	C	C	C																			C	C
Sat	C	C	C	C	C	C	C																	C	C
Sun	C	C	C	C	C	C	C	C																C	C
Legend:																									
C Ramp may be closed completely.																									
REMARKS: Detour traffic via Detour Plan No. 7 on MI-7.																									

Replace section 12-4.02C(3)(m) with:

12-4.02C(3)(m) City Street Closure Hour Charts and City Street Lane Requirement Charts

Comply with the requirements for a city street closures shown in the following charts:

Chart No. M1 City Street Closure																									
Location: Civic Center Dr/ Redwood Hwy									Direction: NB/SB																
Closure limits: Between Merrydale Road and NB On-Ramp from Redwood Highway																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon– Thu	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Fri	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sat	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sun	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Legend:																									
1	Provide at least 1 city street lane open in the direction of travel.																								
REMARKS: The number of through traffic lanes in each direction of travel is 1. Detour pedestrians via Pedestrian Detour Plan Nos. 1 and 2 on MI-10. Access to 5000 Civic Center Dr shall be maintained during use of this chart. Emergency vehicle access shall be provided at all times. This Lane Closure Chart may be used for a maximum of 60 total Working Days.																									

Chart No. <u>M2</u> <u>City Street Closure</u>																									
Location: <u>Civic Center Dr/</u> <u>Redwood Hwy</u>										Direction: <u>NB/SB</u>															
Closure limits: <u>Between Merrydale Road and NB On-Ramp from Redwood Highway</u>																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon– Thu	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>																					
Fri																								<u>1</u>	<u>1</u>
Sat	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>
Sun	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>
Legend:																									
<u>1</u>	Provide at least 1 city street lane open in the direction of travel.																								
REMARKS: The number of through traffic lanes in each direction of travel is <u>1</u> . <u>Detour traffic via Detour Plan Nos. 3, 4, 5, and 6 on Sheets MI-3 through MI-6. Detour pedestrians via Pedestrian Detour Plan Nos. 1 and 2 on MI-10. This Lane Closure Chart may be used for a maximum of 1 occurrence. Access via the Redwood Hwy/Civic Center Dr intersection to 5000 Civic Center Dr shall be closed during use of this chart but shall reopen by 4am Monday.</u>																									

Edits do not require the approval of the specification owner but must be verified by the transportation management plan manager or district traffic manager.

Chart No. M3 Complete City Street Closure																										
Location: Civic Center Dr/ Redwood Hwy									Direction: NB/SB																	
Closure limits: Between Merrydale Road and NB On-Ramp from Redwood Highway																										
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Mon– Thu	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	
Fri	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	
Sat	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	
Sun	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	
Legend:																										
C		Street may be closed.																								
REMARKS: The number of through traffic lanes in each direction of travel is 1. Detour traffic via Detour Plan Nos. 3, 4, 5, and 6 on Sheets MI-3 through MI-6. Detour pedestrians via Pedestrian Detour Plan Nos. 1 and 2 on MI-10. This Lane Closure Chart may be used for a maximum of 60 total Working Days. Access via NB Civic Center Dr to 5000 Civic Center Dr shall be maintained during use of this chart.																										

Chart No. M4 City Street Closure																									
Location: Manuel T Freitas Pkwy										Direction: WB															
Closure limits: Between NB On-Ramp and Northgate Boulevard																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon– Thu	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Fri	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sat	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sun	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Legend:																									
1		Provide at least 1 city street lane open in the direction of travel.																							
REMARKS: The number of through traffic lanes in each direction of travel is 2.																									

Chart No. M5 City Street Closure																									
Location: Manuel T Freitas Pkwy										Direction: EB															
Closure limits: Between Del Presidio Boulevard and NB On-Ramp																									
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Mon– Thu	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Fri	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sat	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Sun	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Legend:																									
1		Provide at least 1 city street lane open in the direction of travel.																							
REMARKS: The number of through traffic lanes in each direction of travel is 2.																									

AA

14 ENVIRONMENTAL STEWARDSHIP

An ESA exists on this project.

Before starting job site activities, install THVF (Temporary High Visibility Fence) to protect the ESA and mark its boundaries.

Install signs 100 feet apart along the length and at the ends of the THVF identifying the area as an ESA. Place the top of each sign flush with the top of the THVF.

The signs must:

1. Be weatherproof and fade-proof
 2. Be from 8-1/2 to 11 inches high and from 11 to 14 inches wide
 3. Have the following message in black letters: Environmentally Sensitive Area – Keep Out
- The message must be legible from a distance of 20 feet by persons with 20/20 vision or vision corrected to 20/20.

The signs may be made of laminated printed paper attached to an inflexible weatherproof backer board.

Attach the signs to the THVF with tie wire or locking plastic fasteners.

Maintain the signs until Contract acceptance.

Limited access to the ESA is allowed for inspection. Notify the Engineer 5 business days or less before the planned entry date. Any other access to the ESA is prohibited.

Access to an ESA other than that described is prohibited.

Add to the 1st paragraph of section 14-6.03A:

This project is within or near habitat for the regulated species shown in the following table:

Regulated Species	
Congested-headed Hayfield Tarplant	
Marin Manzanita	
Pallid Bat	
Burrowing Owl	

This project includes the sensitive habitats shown in the following table:

Sensitive Habitats	
South Fork Gallinas Creek	
Gallinas Creek	

Replace item 1 in the list in the 2nd paragraph of section 14-6.03A with:

1. Stop all work within a 50-foot radius of the discovery except as shown in the following table:

Regulated species	Protective radius (feet)
Active nests of raptors and owls	300
Native birds	100

Add to section 14-6.03A:

Species protection areas within the project limits are as specified in the following table:

Species Protection Areas	
Identification name	Location
Species Protection Area 1 (SPA-1)	Entire project

Within SPA-1, implement the following protection measures:

1. Notify the Engineer at least 30 days before start of construction activities to schedule and attend a biological commitment review meeting with the Engineer and the Department Biologist.
2. Notify the Engineer at least 15 days before the start of construction (e.g., vegetation trimming and/or removal, staging, ESA installation, etc.) to allow for preconstruction surveys to be conducted by the Department Biologist.

3. Notify the Engineer at least 15 days before the start of all vegetation pruning, trimming, or removal to during the nesting bird season (February 1 to September 30) or work on structures (e.g., culverts) suitable for nesting birds to allow for nesting bird surveys to be conducted by the Department Biologist:
 - 3.1. At least 15 days before starting initial work.
 - 3.2. At least 10 days before resuming work.
 - 3.3. Do not start work without authorization.
 - 3.4. Authorization is valid for 72 hours whether or not you perform the work.
 - 3.5. If a nest is discovered, no habitat removal or modification can occur within the buffer until the young have fully fledged.
4. A Department Biologist must be on-site for all vegetation pruning, trimming removal, or any construction activity that may result in a take of special status species, unless otherwise authorized.
5. Before performing any work at the job site, construction personnel must complete a 2-hour environmental training provided by the Department Biologist regarding regulated species, related laws and regulations, and protection measures. Upon completing this training, the construction personnel must sign a form stating that they have completed the training.
 - 5.1. Request training at least 20 days before starting work to coordinate initial training.
 - 5.2. All new personnel, throughout the course of the construction duration, must complete the environmental training before starting work at the job site.
6. Minimize job site disturbance (e.g., vegetation removal) due construction activities.
7. If a special status species is observed on the job site, stop work in the immediate vicinity of said species and notify the Engineer and the Department Biologist.
8. Locate construction staging, storage, and vehicle parking areas on paved or graveled surfaces and outside of Environmental Sensitive Areas (ESA):
 - 8.1. Access routes and the number and size of staging and work areas must be minimized.
 - 8.2. Use of areas not shown on the plans for staging, storage and laydown, vehicle parking, and access must be authorized by the Engineer and Department Biologist.
 - 8.3. Notify the Engineer at least 30 days before requesting staging, storage and laydown, vehicle parking, and access areas not shown on the plans for assessment to be conducted by the Department Biologist.
 - 8.4. If you use any property outside the job site for the sole use of the project, and soil or vegetation will be disturbed:
 - 8.4.1. Obtain all PLACs under applicable environmental laws, regulations, orders, and decrees.
 - 8.4.2. Submit those PLACs at least 15 business days before starting work on the site.
9. Refueling of mobile construction equipment and vehicles must not occur within 100 feet of any water or wetland feature, or anywhere that spilled fuel could drain to a water or wetland feature. Refueling of stationary equipment requiring breakdown and a setup to move will remain in place. Equipment must be refueled with appropriate drip pans, absorbent pads, and water quality Best Management Practices. Equipment and vehicles operating in the project area must be checked and maintained daily to prevent leaks of fuels, lubricants, or other liquids.
10. Do not work in the rain unless authorized. Monitor the weather for forecasted precipitation by National Weather Service. When 1/4 inch or more of precipitation is forecasted to occur, work must stop before precipitation commences. After any storm event, inspect the entire job site for erosion and sediment problems. Notify the Engineer immediately of any problem and take corrective action as

authorized. 72 hour weather forecasts from the National Weather Service must be consulted and work must not start back up until runoff ceases and there is less than a 50 percent forecast for precipitation for the following 24-hour period.

11. Debris and waste must be picked up daily:

11.1. All food and related food trash items must be enclosed in a trash container with a lid and removed from the job site daily.

11.2. Dispose cigarette butts in trash container.

12. All pets are prohibited from the job site.

13. Temporary or permanent erosion control devices containing plastic monofilament netting, including photo-degradable or bio-degradable plastic netting is prohibited.

14. Dust control measures shall be implemented consisting of regular truck watering of construction access areas and disturbed soil areas, and including the use of organic soil stabilizers, to minimize airborne dust and soil particles generated from graded areas. Watering guidelines shall be established to avoid any excessive run-off that may flow into contiguous areas. Any material stockpiles shall be watered, sprayed with tackifier, or covered to minimize dust production and wind erosion.

15. If noxious weeds are disturbed or removed during construction-related activities, contain the plant material associated with these noxious weeds and dispose of them in a manner that will not promote the spread of the species. Obtain all permits, licenses, and environmental clearances for properly disposing of materials. Areas subject to noxious weed removal or disturbance shall be replanted with fast-growing native grasses or a native erosion control seed mixture. If seeding is not possible, the area within the project site shall be covered to the extent practicable with heavy black plastic solarization material until the end of construction.

16. Cut vegetation cut above soil level, except in areas that will be excavated. This will allow plants that to resprout after construction. Clear and grub woody vegetation by hand or use equipment, such as mowers, backhoes, and excavators. Remove cleared vegetation from the project footprint to prevent the attraction of animals to the project site.

17. All excavated steep-walled holes or trenches more than 1-foot deep must be covered at the end of each working day by plywood or similar materials or provide with one or more escape ramps constructed of earthen fill or wooden planks at an angle no greater than 30 degrees.

17.1. Before all excavated holes or trenches are filled, they must be thoroughly inspected for trapped animals.

17.2. Stop work and immediately notify the Engineer and Department Biologist if trapped wildlife is discovered.

17.3. Do not resume activities until authorized.

18. Notify Engineer and obtain permission 5 days before entering unfenced (THVF) ESA areas.

19. Notify Engineer 10 days before establishing staging and storage areas within SPA_1 for approval.

Add after the 2nd paragraph of section 14-11.12A:

This project includes removal of yellow thermoplastic traffic stripe that will produce hazardous waste residue.

Add after the 1st paragraph of 14-11.12E:

After the Engineer accepts the analytical test results, dispose of yellow thermoplastic and yellow paint hazardous waste residue at a Class 1 disposal facility located in California 90 days after accumulating 220 lb of residue.

If less than 220 lb of hazardous waste residue and dust is generated in total, dispose of it within 60 days after the start of accumulation of the residue.

Replace section 14-11.14 with:

14-11.14 TREATED WOOD WASTE

14-11.14A General

Section 14-11.14 applies if treated wood waste is shown on the Bid Item List.

Section 14-11.14 includes specifications for handling, storing, transporting, and disposing of treated wood waste. Manage treated wood waste under Health & Safety Code §25230 et seq.

Wood removed from roadside signs is treated wood waste.

14-11.14B Submittals

Within 5 business days of disposing of treated wood waste, submit as an informational submittal a copy of each completed shipping record and weight receipt.

14-11.14C Training

Provide training to personnel who handle or may come in contact with treated wood waste. Training must include:

1. Requirements of 8 CA Code of Regs
2. Procedures for identifying and segregating treated wood waste
3. Safe handling practices
4. Requirements of Health & Safety Code §25230 et seq
5. Proper disposal methods

Maintain training records for 3 years after contract acceptance.

14-11.14D Storage of Treated Wood Waste

Store treated wood waste at the jobsite until transport to the CA permitted disposal site.

Until disposal, store treated wood waste using the following methods:

1. Raise the waste on blocks above a foreseeable run-on elevation and protect it from precipitation for no more than 90 days.
2. Place the waste on a containment surface or pad protected from run-on and precipitation for no more than 180 days.
3. Place the waste in water-resistant containers designed for shipping or solid waste collection for no more than 1 year.
4. Place the waste in a storage building as defined in Health & Safety Code §25230 et seq.

Prevent unauthorized access to treated wood waste using a secure enclosure such as a locked chain-link-fenced area or a lockable shipping container located within the job site.

Resize and segregate treated wood waste at a location where debris including sawdust and chips can be contained. Collect and manage the debris as treated wood waste.

Identify treated wood waste and accumulation areas using water-resistant labels that comply with Health & Safety Code §25230 et seq. Labels must include:

1. The words *TREATED WOOD WASTE Do not burn or scavenge*
2. The words *Caltrans District* and the district number
3. The words *Construction Contract* and the contract number
4. District office address
5. Engineer's name, address, and telephone number
6. Contractor's contact name, address, and telephone number
7. Date placed in storage

Dispose of treated wood waste within:

- Before transporting treated wood waste, obtain agreement from the receiving facility that it will accept the waste. Protect shipments of the waste from loss and exposure to precipitation. For projects generating 10,000 lb or more of treated wood waste, request a generator's EPA Identification Number from the Engineer at least 5 business days before the 1st shipment. Each shipment must be accompanied by a shipping record such as a bill of lading or invoice that includes:

- The shipping record must be 8-1/2 by 11 inches and a 4-part carbon or carbonless form to provide copies for the Engineer, transporter, and treated wood waste facility.

Dispose of treated wood waste at one of the following:

- [illegible]

Contract No. 04-0K8004

Pervious backfill material placed within the limits of payment for bridges is paid for as structure backfill (bridge). Pervious backfill material placed within the limits of payment for retaining walls is paid for as structure backfill (retaining wall).

20 LANDSCAPE

Hand pull weeds within biofiltration swale and strip areas.

Control weeds under guard rails, from within asphalt concrete surfacing, concrete surfacing, rock blankets, and unpaved gore areas between the edge of pavement and planting areas with pesticides or by hand pulling.

Remove by hand pulling or mechanical means noxious and invasive plants within the highway. Noxious and invasive plants are shown in the table below:

Botanical name	Common name	Cover seed pods/heads
<i>Acacia dealbata</i>	Silver Wattle	Yes
<i>Acacia melanoxylon</i>	Black Acacia	Yes
<i>Cotoneaster lacteus</i>	Parney Cotoneaster	Yes
<i>Cortaderia jubata</i>	Jubata Grass	Yes
<i>Cortaderia sellowana</i>	Pampas Grass	Yes
<i>Echium candicans</i>	Pride of Madeira	Yes
<i>Eucalyptus globulus</i>	Blue Gum	Yes
<i>Foeniculum vulgare</i>	Fennel	Yes
<i>Myoporum laetum</i>	Myoporum	Yes
<i>Nassella tenuissima</i>	Mexican Feather Grass	Yes
<i>Pinus radiata</i>	Monterey Pine	Yes
<i>Prunus cerasifera</i>	Cherry Plum	Yes
<i>Pyracantha coccinea</i>	Firethorn	Yes

Spot spray with pesticides and remove noxious and invasive plants, roots, and root crowns within the highway. Noxious and invasive plants are shown in the table below:

Botanical name	Common name	Cover seed pods/heads
<i>Acacia dealbata</i>	Silver Wattle	Yes
<i>Acacia melanoxylon</i>	Black Acacia	Yes
<i>Cotoneaster lacteus</i>	Parney Cotoneaster	Yes
<i>Cortaderia jubata</i>	Jubata Grass	Yes
<i>Cortaderia sellowana</i>	Pampas Grass	Yes
<i>Echium candicans</i>	Pride of Madeira	Yes
<i>Eucalyptus globulus</i>	Blue Gum	Yes
<i>Foeniculum vulgare</i>	Fennel	Yes
<i>Myoporum laetum</i>	Myoporum	Yes
<i>Nassella tenuissima</i>	Mexican Feather Grass	Yes
<i>Pinus radiata</i>	Monterey Pine	Yes
<i>Prunus cerasifera</i>	Cherry Plum	Yes
<i>Pyracantha coccinea</i>	Firethorn	Yes

For information on photographic identification of weed species go to the Department of Food and Agriculture or the California Invasive Plant Council websites.

Seed head or pod removal is shown in the table above. Securely cover seed heads or pods with plastic and remove and dispose. Prevent seed dispersal during transportation to the disposal site.

Replace the 2nd paragraph of section 20-1.03C(4) with:

Dispose of mowed material during roadside clearing. Dispose of noxious and invasive plants within 3 days of removal. Dispose of seed pods and heads the same day as removed. Prevent seed dispersal during transportation to the disposal site.

Replace the 1st paragraph of section 20-2.01B(7) with:

Valve boxes must be polymer concrete.

Add to the 1st paragraph of section 20-2.01B(7):

Valve boxes must not have side openings.

Replace item 1 in the list in the 2nd paragraph of section 20-2.01B(7) with:

1. Polymer concrete

Add to the list in the 2nd paragraph of section 20-2.01B(7):

4. Secured using a commercial-quality locking mechanism

Add between the 4th and 5th paragraphs of section 20-2.01B(7):

Remote control valves must be labeled with a polyurethane tag. Attach the tag tightly with a nylon tie to the conductor wire. The tag must be stamped on both sides with the appropriate letters and numbers at least 1 inch high showing the valve's controller and station.

Add to section 20-2.05B:

Flow sensor cable must:

1. Be rated for 600V.
2. Be rated for 194 degrees F.
3. Be UL listed as Type TC.
4. Comply with specifications of ICEA/NEMA.
5. Consist of two no. 16 minimum stranded copper conductors. Insulated conductor must be color coded with a PVC or nylon jacket.
6. Include a tinned copper braid or aluminized polyester film shield. Where the film is used, a no. 18 or larger, stranded or no. 16 solid, tinned, copper drain wire must be placed between the insulated conductors and the shield and in contact with the conductive surface of the shield.
7. Include a black PVC jacket with a minimum nominal thickness of either 50 mils or 48 mils where capacitance of conductors to other conductors and the shield is 87 pF/ft or better. The cable jacket must be marked with the insulation type designation, conductor size, and voltage and temperature ratings.
8. Have an outside diameter from 0.29 to 0.45 inch.
9. Be UV resistant and direct burial type.
10. Have no splices between components except where shown.

Add to the list in the 1st paragraph of section 20-2.06B(2)(a):

17. Be EPA WaterSense® approved.

Add to section 20-2.06B(2)(a):

The irrigation controllers within Department highway areas must be WeatherTrak LC+ and must have 2-way communication by 5G cellular service. The vendor must install any necessary software and conduct any initial software or proprietary website setup configuration for communications between controller and any web-enabled device.

You may obtain specified equipment listed below from:

Company: WATERSAVERS IRRIGATION, Inc
Address: 4306 Redwood Highway Suite 200, San Rafael, CA 94901
Business phone number: (415) 256-1711
Email address: sanrafaelstore@watersaversinc.com

The Department has obtained quoted prices, not including sales tax and delivery, for the equipment shown in the following table:

Equipment description	Quoted price (EA)	Quantity	Extended price	Controller and Remote Access Device identification
Irrigation Controller	\$1,264	1	\$1,264	WTLCC06PL
Pedestal subassembly	\$1,050	1	\$1,050	WTLCCPED
Flow management & monitoring	\$315	1	\$315	WTLCFLOKEY
Communication service (4-year plan extension)	\$956	1	\$956	CIM4Y

These prices are good until 07/01/2024.

Delete items 2.1, 2.2, and 2.3 in the list in the 1st paragraph of section 20-2.06B(3).

Add to section 20-2.06B(3):

A single irrigation controller enclosure cabinet must be 48 inches high by 18 inches wide by 12 inches deep.

Replace the paragraph of section 20-2.10B(2) with:

Each ball valve must be PVC or chlorinated PVC and must comply with the requirements shown in the following table:

Quality characteristic	Requirement
Nonshock working pressure for 3/4 to 4 inch valves (min, psi)	235
Nonshock working pressure for 6 inch valves (min, psi)	150
Seats	PTFE
O-ring seals	EPDM or fluoroelastomer

Replace item 4 in the 1st paragraph of section 20-2.10B(4) with:

4. Pressure regulator that:
 - 4.1. Can withstand a working pressure of 150 psi
 - 4.2. Has a factory preset outlet pressure of 30 to 40 psi; printed on the exterior surface
 - 4.3. Can be installed above or below grade

Add to section 20-2.10B(8):

Pressure relief valves must be factory set at 75 psi.

Replace the 1st paragraph of section 20-2.10B(9) with:

Each quick coupling valve connected to recycled water must be 1-inch acme thread key with a non-potable locking purple thermoplastic rubber cap, 1-inch brass key and 1-inch brass hose swivel. Except for the cap, quick coupling valve must be brass or bronze construction.

Add to the list in 1st paragraph of section 20-2.10B(10)(a):

12. Be equipped with a self-flushing feature for recycled water.
13. Not have external tubing.

Add to the end of section 20-2.10B(10)(a):

Each remote control valve must have:

1. Chlorine and chemical -resistant diaphragm and scrubber components
2. Internal scrubber mechanism
3. Purple colored manual flow control handle
4. Recycled water warning tag

Add to section 20-3.01A(3)(b):

Some plants required may not be readily available and must be grown specifically for this project. Submit a statement within 30 days after Contract approval from the vendor that the order to grow the plants, including inspection plants and replacement plants, has been received and accepted by the vendor. The statement from the vendor must include the plant names, sizes, and quantities and the anticipated delivery date. Notify the Engineer when the vendor has started growing the plants.

Add to section 20-3.01B(3)(a):

Soil amendment must be fine or medium particle size compost, or a mixture of both, under section 21.

Add to section 20-4.01A:

This project has a Type 2 plant establishment period.

Replace the 1st paragraph of section 20-4.01C(1) with:

Submit the following seasonal watering schedules for use during the plant establishment period:

1. March through May
2. June through August
3. September through October
4. November through February

Submit the first season's watering schedule within 10 days after the start of the plant establishment period. Submit subsequent watering schedules at least 5 business days before start of the next seasonal period. Remote irrigation control system watering schedule must use the remote irrigation control system software program.

Add to the beginning of the 1st paragraph of section 20-4.03A:

Maintain a neat and presentable job site during plant establishment including areas not visible to the public.

Apply organic fertilizer to the plants during the 1st week of March and September of each year.

Dispose of weeds under section 20-1.03C(4).

Control pests under sections 20-1

Do not use preemergent.

AA

Process compost materials under 14 CA Code of Regs § 17868.3.

The particle size must comply with the requirements shown in the following table:

Compost Gradation

Quality characteristic	Test method ^a	Requirement	
		Min	Max
Gradation Fine:(dry weight % passing) 1-inch sieve 3/8-inch sieve	TMECC 02.02-B	99 95	-- --
Gradation Medium:(dry weight % passing) 2-inch sieve 3/8-inch sieve	TMECC 02.02-B	95 30	-- 75
Gradation Coarse:(dry weight % passing) 3-inch sieve 3/8-inch sieve	TMECC 02.02-B	95 --	-- 50

^aTMECC refers to *Test Methods for the Examination of Composting and Compost*, published by the United States Department of Agriculture and the United States Compost Council (USCC).

The quality characteristics of compost must have the values shown in the following table:

Compost

Quality characteristic	Test method ^a	Requirement		
		Fine	Medium	Coarse
pH	TMECC 04.11-A	6.0–8.5	6.0-8.5	5.0-8.5
Soluble salts (dS/m)	TMECC 04.10-A	0–10	0-10	0-10
Moisture content (% wet weight)	TMECC 03.09-A	25–60	25-60	< 60
Organic matter content (% dry weight)	TMECC 05.07-A	30–70	30-100	25-100
Maturity (seed emergence) (% relative to positive control)	TMECC 05.05-A	80 or above	80 or above	N/A
Maturity (seedling vigor) (% relative to positive control)	TMECC 05.05-A	80 or above	80 or above	N/A
Stability (mg CO ₂ -C/g OM per day)	TMECC 05.08-B	5 or below	8 or below	8 or below
Pathogen Salmonella (most probable number per 4 grams dry weight basis)	TMECC 07.01-B	< 3	< 3	< 3
Pathogen Fecal coliform (most probable number per gram dry weight basis)	TMECC 07.01-B	< 1,000	< 1,000	< 1,000
Physical contaminants (% dry weight) Plastic, glass, and metal	TMECC 02.02-C	combined total: < 0.5	combined total: < 1.0	combined total: < 0.5
Film plastic (% dry weight)	TMECC 02.02-C	Combined total: < 0.1	Combined total: < 0.1	Combined total: < 0.1

^a TMECC refers to *Test Methods for the Examination of Composting and Compost*, published by the United States Department of Agriculture and the United States Compost Council (USCC).

Add to the end of section 21-2.04:

The following bid items are eligible for Move-in/Move-out (Erosion Control) payments:

- [illegible]

39 ASPHALT CONCRETE

[illegible]

A A

63

Bridge no./Structure name	Description of work
RW1 Retaining Wall	Portion of retaining wall as shown.
RW2 Retaining Wall	Portion of retaining wall as shown.

AA

DIVISION VIII MISCELLANEOUS CONSTRUCTION

73 CONCRETE CURBS AND SIDEWALKS

Add to section 73-1.02A:

Concrete must be minor concrete complying with section 90-2 and may contain returned plastic concrete complying with section 90-9.

Add to section 73-3.01C:

Within 2 business days of completing the surveys, submit preconstruction and post-construction surveys sealed and signed by one of the following:

1. Land surveyor licensed in the State
2. Engineer who is registered as a civil engineer in the State

Replace section 73-3.01D(3) with:

73-3.01D(3) Quality Control

For locations shown, perform a preconstruction survey to ensure forms and job site constraints will allow for compliance with required design dimensions and slopes shown. Upon completing the work, perform a post-construction survey to verify design dimensions and slopes requirements are met. The post-construction survey must include a minimum of 3 measurements for each dimension and slope requirement shown. Individual measurements must be equally distributed across the specified slope or dimensional surface. Document and submit these measurements on the Americans with Disabilities Act Compliance Inspection Report form for the facility type shown. Include the equipment and control used to conduct the survey.

Add to the beginning of section 73-3.03:

Before placing concrete, verify that forms and job site constraints allow the required dimensioning and slopes shown. Immediately notify the Engineer if you encounter job site conditions that will not accommodate the design details. Ordered modifications are change order work.

AA

78 INCIDENTAL CONSTRUCTION

Add to the end of section 78-4.04B(2)(a):

The stain must be a water-based stain.

The final color of the concrete surface texture wood plank must match color shown. The final color of the class 1 finish must match color shown

78-24 RELOCATE BUS SHELTER AND TRASH RECEPTACLE

[illegible]

2. The daytime luminance factor (Y), for green route shield pavement markings shall be at least 6, but no more than 15. Color chromaticity limits for green route shield pavement markings at daytime must plot within the boundaries shown in the following table:

Daytime Chromaticity Coordinates for Green Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.260	0.399	0.166	0.364	0.286	0.446	0.207	0.771

3. Color chromaticity limits for green route shield markings at nighttime must plot within the boundaries shown in the following table:

Nighttime Chromaticity Coordinates for Green Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.007	0.570	0.200	0.500	0.322	0.590	0.193	0.782

4. The daytime luminance factor (Y), for red route shield pavement markings, shall be at least 6, but no more than 15. Color chromaticity limits for red route shield pavement markings at daytime must plot within the boundaries shown in the following table:

Daytime Chromaticity Coordinates for Red Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.648	0.351	0.735	0.265	0.629	0.281	0.565	0.346

5. Color chromaticity limits for red route shield pavement markings at nighttime must plot within the boundaries shown in the following table:

Nighttime Chromaticity Coordinates for Red Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.650	0.348	0.620	0.348	0.712	0.255	0.735	0.265

6. The daytime luminance factor (Y), for blue route shield pavement markings, shall be at least 5, but no more than 14. Color chromaticity limits for blue route shield pavement markings at daytime must plot within the boundaries shown in the following table:

Daytime Chromaticity Coordinates for Blue Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.078	0.171	0.150	0.220	0.210	0.160	0.137	0.038

7. Color chromaticity limits for blue route shield pavement markings at nighttime must plot within the boundaries shown in the following table:

Nighttime Chromaticity Coordinates for Blue Route Shield Pavement Marking (Corner Points)

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
0.033	0.370	0.180	0.370	0.230	0.240	0.091	0.133

Replace section 84-6 with:

84-6 TRAFFIC STRIPE TAPE WARRANTY

84-6.01 GENERAL

84-6.01A Summary

Section 84-6 includes the warranty requirements for traffic stripe tape.

Traffic stripe tape must comply with section 84-2.

Traffic stripe tape must be measured under the test methods and frequencies shown in the following table:

Test Methods and Frequencies for Traffic Stripes

Quality characteristic	Test method	Minimum sampling and testing frequency	Requirement
Durability (min, %)	ASTM D913	Visual	100
Initial retroreflectivity (min, $\text{mcd} \cdot \text{m}^{-2} \cdot \text{lx}^{-1}$)	ASTM E1710	ASTM D7585 ^{a,b}	700 500
White			
Yellow			
Color ((x,y) chromaticity coordinates)	ASTM D6628	Per lot number	Table 1 Table 2
Daytime			
Nighttime			

^aUse the referee evaluation protocol for project length less than 10 miles. For project lengths greater than or equal to 10 miles, add one evaluation for every additional mile.

^bMeasure retroreflectivity at least 48 hours after placement and within 30 days of applying the traffic stripes.

84-6.01B Definitions

Not Used

84-6.01C Submittals

84-6.01C(1) General

Twenty five days before placing the traffic stripe tape, submit to the Engineer and the Division of Maintenance:

1. Contractor's traffic stripe tape warranty for durability, color, and retroreflectivity
2. Completed Warranty Bond form (TOTE-1)
3. Name of the manufacturer's representative or name and certificate of the manufacturer certified contractor, who will monitor the installation

Before contract acceptance, submit in electronic format to maintenance.striping.warranty.contact@dot.ca.gov and to the Engineer, in an authorized data-storage device, the following information:

1. Project identification number
2. Project location information, including:
 - 2.1. District

- 2.2. County
- 2.3. Route
- 3. Stripe information, including:
 - 3.1. Standard plan detail number
 - 3.2. Contrast as *y* or *n*
 - 3.3. Date installed as mm/dd/yyyy
 - 3.4. Initial retroreflectivity number to 1 decimal place

During the warranty period:

- 1. Within 20 days of receiving notification from the Division of Maintenance that traffic stripes are deficient, submit a traffic stripe replacement plan and schedule to the Division of Maintenance
- 2. Within 5 days of installation of the replacement traffic stripe tape, submit the retroreflectivity and color test data

84-6.01D Quality Assurance

84-6.01D(1) General

Not Used

84-6.01D(2) Warranty

84-6.01D(2)(a) General

The warranty period:

- 1. For traffic stripe tape is 4 years
- 2. Starts the day after Construction Contract Acceptance (CCA date)

The warranty bond must be equal to 100 percent of the total payment for the bid items subject to the warranty. The bond must be in effect for the entire warranty period, including the time to perform corrective work. Each bond must be provided by a surety licensed to do business in the State.

You are responsible for the costs of removing and replacing traffic stripes that are noncompliant with the performance requirements during the warranty period. These costs include, but are not limited to, surface preparation, material, equipment, labor, encroachment permit fees, and traffic control. All warranty work must be performed at no cost to the Department. The replacement materials will only be covered by the remainder of the original warranty period.

The warranty does not cover damages due to acts of God.

84-6.01D(3) Quality Control

Traffic stripes must maintain the performance requirements throughout the warranty period as follows:

- 1. Daytime and nighttime color chromaticity coordinates
- 2. Retained retroreflectivity of $175 \text{ mcd} \cdot \text{m}^{-2} \cdot \text{lx}^{-1}$
- 3. Minimum 90 percent durability rating for any 400-foot segment

84-6.02 MATERIALS

Traffic stripe tape must be on the authorized material list for signing and delineation materials.

Traffic stripe tape must have a precoated, pressure-sensitive adhesive.

Traffic stripe tape must have an embossed pattern of raised surfaces.

During the warranty period, the replacement materials must comply with section 84-2 and meet or exceed the performance of the original materials.

84-6.03 CONSTRUCTION

A manufacturer's representative must be present during the installation or a manufacturer-certified contractor must install the traffic stripe tape.

During the warranty period:

An equipment grounding conductor must be bare.

The individual conductors in the cable must be solid copper complying with ASTM B3 with Type THWN insulation. The minimum thickness of insulation must comply with NEC for conductor sizes no. 14 to no.10. The minimum thickness of the nylon jacket must be 4 mils.

1. Be 3 to 7 mils thick
2. Have black capital character on a white background extending at least 1/4 inch beyond the character
3. Be clear surface coated 2 to 3 mils thick with translucent luster ultraviolet protectant
4. Use permanent adhesive
5. Last for at least 7-year without fading and peeling

The interior of the enclosure must accept cable-in/cable-out circuit breakers. The circuit breakers must be mounted on nonenergized clips and vertically with the up position of the handle being the *ON* position.

For service equipment enclosure, type III-A series, provide a minimum clearance of 24 inches between the bottom of the lowest circuit breaker and the bottom of the enclosure

4. Be made of metal

[illegible]

1. The Engineer
2. Department's Electrical and Signal Maintenance Superintendent at (415) 330-6500

The shutdown of traffic signal systems is allowed only between the hours of 9 pm and 5am.

Add between the 23rd and 24th paragraphs of section 87-1.03A:

Where a Type A or a Type B loop detector is shown, a Type E loop detector may be substituted. Use only one type of loop detector per system.

Where a Type D loop detector is shown, a Type F loop detector may be substituted. Use only one type of loop detector per system.

Add between the 23rd and 24th paragraphs of section 87-1.03A:

When replacing inductive detector loops, splice loops to the detector lead-in cable (DLC) and match loop designations with the DLC labels inside the controller cabinet accordingly.

Perform conductor test for the detector lead-in cable (DLC) at the controller cabinet before connecting DLC to the terminal block (1) after disconnecting existing loops and DLC in the termination pull box and (2) after connecting the newly installed loop to the DLC. Submit test results:

1. 1 day before starting loop replacement at each location
2. 1 day after replacement at all locations is completed

Replace the 20th paragraph of section 87-1.03B(1) with:

Terminate conduit through the bottom of a nonmetallic pull box 2 inches above the bottom and 2 inches from the wall closest to the direction of the run.

Add to the end of section 87-1.03B(1):

Where 6 or more 3-inch conduit enter a no. 6 pull box, the conduit must enter at an angle not greater than 45 degrees from the horizontal.

Add to the beginning of section 87-1.03B(3)(a):

Use Type 3 conduit for underground installation.

Use Type 3, Schedule 80 conduit in a foundation and between a foundation and the nearest pull box.

Replace the 3rd paragraph of section 87-1.03C(2)(a) with:

Install a pull box on a bed of crushed rock.

Replace the 1st paragraph of section 87-1.03F(2)(c)(ii) with:

Install a Type B loop detector lead-in cable in conduit.

Replace the 1st paragraph of section 87-1.03F(3)(c)(ii) with:

Use a Type 2 loop wire. Use only Type 2 loop wire for Type E and F loop detectors.

Replace the 2nd paragraph of section 87-1.03H(2) with:

Use Method B to insulate a splice.

Add to the end of section 87-1.03L(2)(b):

Install a Type III service equipment enclosure for irrigation controllers 12 STATION.

Replace the 2nd paragraph of section 87-1.03E(3) with:

Construct a concrete pad in front of a Type III service equipment enclosure. The pad must be 36 inches long, 30 inches wide, and 4 inches thick.

Replace the 4th paragraph of section 87-1.03T with:

Connect the signal interface cable to the APS and the terminal block in the pedestrian signal head.

Add to the end section 87-1.03T:

A manufacturer's representative must program the accessible pedestrian signals at the following intersections:

1. Intersection of 101SB Off ramp/Del Presidio Blvd and Manual T Freitas Pkway

When the extended pushbutton press is used, program the signals with messages for each street as follows:

1. During the pedestrian clearance interval, the message heard must be *Wait to Cross Manual T Freitas Pkwy. Wait.*
2. During the pedestrian clearance interval, the message heard must be *Wait to Cross 101 SB Off ramp. Wait.*
3. During the pedestrian clearance interval, the message heard must be *Wait to Cross Del Presidio Blvd. Wait.*

Provide training by a representative certified by the manufacturer for up to 30 Department employees selected by the Engineer. Provide all materials and equipment for the training. Notify the Engineer 20 days in advance of the proposed training to obtain authorization for the time and place of the training. If agreement cannot be reached, the Engineer determines the time and place.

Add to the end of section 87-1.03V(1):

Where 1 or more traffic signal detectors consist of a sequence of 4 loops in a single lane, locate the front loop closest to limit line or crosswalk 1 foot from the line. Connect the set of 3 or 4 loops assigned to the same loop detector lead-in cable (DLC) in series for traffic signal system.

Add between the 9th and 10th paragraphs of section 87-1.03V(2):

Use elastomeric sealant or hot-melt rubberized sealant to fill slots.

Add to the end of section 87-21.03C:

Modifying a lighting system includes removing, adjusting, or adding:

1. Foundations
2. Pull boxes
3. Conduit
4. Conductors
5. Standards
6. Luminaires
7. Service equipment enclosure
8. Photoelectric control

